

Tariff Cuts, External Shocks, and Strategic Inputs: The EU–Mercosur Agreement in a Changing World

Guilherme Paiva Pinto*

July 16, 2026

Abstract

The EU–Mercosur agreement reached final signature in a different economic setting from the one that surrounded the early negotiations. Talks opened in the early 2000s, but the final goods-tariff schedule was signed only in 2026, after Mercosur’s trade with China deepened, U.S. tariff risk became more salient, and strategic-input concerns moved closer to the center of industrial-policy debate. I ask whether these changes make the agreement easier to defend on economic grounds, even when direct goods-tariff gains are small. The exercise codes the 2026 Annex 2-A product-level tariff-elimination schedule and uses it as a common counterfactual shock in the 2007, 2011, 2014, 2017, and 2022 benchmark economies using a six-region, 49-sector Caliendo–Parro model. In 2022, the schedule raises welfare by 0.009 percent for Argentina, 0.016 percent for Brazil, and 0.005 percent for the European Union. These aggregate effects are small, but the timing of the agreement also changes where losses are concentrated, how much U.S. tariff risk it offsets in selected EU sectors, and how its value compares with rare-earth input-cost exposure. The paper therefore treats the agreement as a timing problem: the tariff component is modest, but the economic environment in which it is evaluated can make the final signed schedule more defensible than earlier benchmark economies would suggest.

1 Introduction

The EU–Mercosur agreement reached signature after more than twenty years of negotiation. I ask whether the changed economic environment altered the economic case for the final goods-tariff schedule. I use the signed 2026 schedule as a common counterfactual benchmark, not as a claim that the same text was available in every earlier year. The exercise asks how the final schedule would perform when introduced into different benchmark economies, and it compares that result with full-liberalization tariff benchmarks. Long-delayed trade agreements are often discussed as if their value were fixed at the moment of negotiation, even though trade patterns, outside risks, sectoral exposure, and hedging motives can change before implementation. The exercises show that the direct tariff component is economically small, especially in aggregate welfare terms, but the timing of implementation changes the interpretation: sectoral losses remain politically relevant, U.S. tariff risk creates selected offset margins, and rare-earth input-cost exposure can be larger than the direct goods-tariff effect.

The counterfactual comparison holds the 2026 legal tariff-cut schedule constant while the benchmark economy changes across years. This isolates the economic environment from the legal object being changed. The effects of tariff cuts depend on the economy in which they are introduced. Over the years in which the agreement remained unresolved, Mercosur’s trade orientation shifted, China became a more important outside partner, U.S. trade policy became a larger source of risk, and critical inputs became more central to industrial policy. These changes enter the model through trade shares, sector sizes, input-output links, and outside sourcing options. In that sense, the agreement was negotiated in one environment and signed in another.

The agreement was contested through concrete sectoral margins: Mercosur manufacturing adjustment, EU

*Department of Economics, Indiana University.

agricultural exposure, environmental enforcement, procurement rules, and automotive liberalization all shaped the politics of the agreement. Customs-union theory suggests that preferential agreements can raise aggregate welfare while placing losses in specific sectors or members. The paper therefore reports average real-income effects together with the location and concentration of gains and losses.

Annex 2-A is the official document that supplies the measurable goods-tariff shock used to build the benchmark counterfactual. It is a product-level tariff-elimination schedule for goods in the EU–Mercosur interim Trade Agreement and Partnership Agreement ([European Commission, 2026b](#); [European Union and Mercosur, 2026a](#)). I map those product-level commitments into sectoral tariff changes for Argentina, Brazil, and the European Union. This benchmark is intentionally narrower than the full agreement. Services, procurement, quota rents, rules of origin, safeguards, sustainability enforcement, and dispute settlement remain part of the institutional setting outside the direct model shocks. Rare-earth-related products also enter Annex 2-A only as ordinary goods lines, not as a separate strategic-minerals chapter.

In the model, a tariff cut changes delivered prices and sourcing, which in turn changes production costs through input-output links. I use a multi-country, multi-sector trade model with input-output linkages, following [Caliendo and Parro \(2015\)](#), to keep these margins in one equilibrium system. The model has six regions: Argentina, Brazil, China, the European Union, the United States, and the rest of the world. It solves counterfactuals in 49 sectors for 2007, 2011, 2014, 2017, and 2022. OECD ICIO provides the production network ([Organisation for Economic Co-operation and Development, 2023](#)), BACI provides product-level trade weights ([Gaulier and Zignago, 2010](#)), MACMAP provides applied tariffs ([Guimbard et al., 2012](#)), and WITS/UNCTAD TRAINS tariff variation supports the elasticity updates ([World Bank and UNCTAD, 2026](#)). Using the 2026 legal schedule as a common shock while changing the benchmark economy makes the timing question measurable.

Two additional scenarios capture margins that became more salient during the negotiation delay. The first is a U.S. tariff-risk scenario. It applies a 10 percentage-point increase in U.S. goods-import tariffs in the 2022 economy and asks whether Annex 2-A offsets sectoral losses from that external shock. Official U.S. sources define the policy environment used to motivate the stress test ([Federal Register, 2025](#); [The White House, 2025b](#)). The second is a rare-earth minerals exposure scenario. It introduces an EU input-cost reduction for rare-earth-related inputs, interpreted as insurance against export-tax and export-restriction price premia, and an Argentina–Brazil upstream production-cost check. The mechanism is motivated by production-network propagation and rare-earth supply-chain risk ([Baqae and Farhi, 2022](#); [Alfaro et al., 2025](#)). I report these as external-margin exercises around Annex 2-A; they show how external risk changes the interpretation of the tariff schedule.

The small welfare effects guide the interpretation: the tariff schedule is relatively narrow to explain the renewed economic case on its own. Under Annex 2-A, Argentina’s welfare gain is largest in 2007, at 0.031 percent, and falls to 0.009 percent in 2022. Argentina has a small welfare loss of 0.0009 percent in 2011. Brazil’s gain ranges from 0.016 to 0.027 percent, while the European Union gains less than 0.010 percent in every benchmark year. These magnitudes do not support a story in which the goods-tariff schedule alone explains the renewed economic case for the agreement. They instead tell us that the interpretation must come from the distribution of losses and from other external margins that changed during the delay.

Because the aggregate gains are relatively low, the distribution of sectoral gains and losses carries much of the economic interpretation. Aggregate gains coexist with two sectoral resistance margins: concentrated manufacturing losses in Argentina and Brazil, and EU agri-food exposure. The Argentina–Brazil manufacturing concern becomes smaller in levels but does not disappear. In 2022, the output share-weighted legacy manufacturing loss share is 27.1 percent in Argentina and 30.2 percent in Brazil. The EU agri-food margin is small at the bloc level, but remains politically salient because member-state and product-level exposure is hidden inside the aggregate EU region. This is the sense in which small welfare effects can still matter for the politics of approval.

One of the changes I quantify here happens when the external environment becomes more protectionist. I use the U.S. tariff-risk scenario to show how a more protectionist outside environment changes the interpretation of the

agreement's gains. In the 2022 counterfactual, Annex 2-A raises Argentina's imports from the European Union by 7.6 percent and Brazil's by 9.0 percent. When U.S. import tariffs rise in the stress test, the agreement partly offsets sectoral output losses in selected EU mining and manufacturing sectors, but not in Argentina or Brazil. The scenario identifies selective sectoral insurance in a more difficult outside environment.

By comparing the tariff schedule with input-cost exposure, the rare-earth minerals exercise shows where tariffs stop being the main economic margin. In 2022, Annex 2-A alone raises EU welfare by 0.005 percent. Annex 2-A plus the EU rare-earth input-cost scenario raises it by 0.105 percent. Placing the primitive on Argentina–Brazil upstream production instead shifts the welfare gains toward Argentina and Brazil. I report the rare-earth results separately because input-cost placement changes the country pattern of welfare effects.

Because the object being measured is narrower, these estimates are smaller than deep-agreement estimates. For example, [Timini and Viani \(2020\)](#) evaluate a broader EU–Mercosur-style trade-cost shock using a structural-gravity approach. I change the coded legal goods-tariff schedule. The gap in magnitudes reflects the narrower estimand used here. The separate full-liberalization benchmark, which sets bilateral goods tariffs to zero, gives only 0.025 percent for Argentina, 0.037 percent for Brazil, and 0.009 percent for the European Union in 2022. Larger effects must come from non-tariff, deep-agreement, external-policy, or strategic-input channels outside the benchmark tariff schedule.

The contribution is a decomposition of a long-delayed agreement into measurable components: the final goods-tariff schedule, its sectoral distribution, and two external exposure margins. The paper codes the signed legal goods-tariff schedule, evaluates that schedule as a counterfactual object across repeated benchmark years, reports country-sector distribution, and compares the tariff component with zero-tariff full liberalization, U.S. tariff-risk, and rare-earth input-cost scenarios. This design connects the customs-union concern with distribution to the quantitative trade-policy practice of counterfactual analysis, while keeping the legal tariff schedule separate from broader agreement provisions. Section 2 describes the agreement and the political margins that motivate the exercise. Section 3 presents the mechanism, model environment, and measured margins. Section 4 describes the data, tariff construction, and calibration. Section 5 presents the results. Section 6 concludes.

Related Literature. The first literature thread explains why customs unions and preferential agreements can create aggregate gains while distributing costs unevenly across members and sectors. The starting point is the trade-creation and trade-diversion distinction in [Viner \(1982\)](#). [Meade \(1955\)](#) places customs-union effects in a general-equilibrium setting, and [Lipsey \(1957\)](#) shows why welfare depends on both production and consumption margins. [Kemp and Wan \(1976\)](#) provides a welfare-improving benchmark with appropriate common external tariffs and transfers. These papers matter here because a customs union can create aggregate gains while leaving members and sectors with very different interests. [Riezman \(1985\)](#) links this problem to the distribution of gains inside the union. Political-economy models then explain why concentrated sectors can matter for agreement support: [Krishna \(1998\)](#) shows how lobbying can favor trade-diverting preferential agreements, and [Grossman and Helpman \(1994\)](#) provides the standard protection-for-sale logic. I use this literature to interpret concentrated losses as pressure margins, without estimating votes, lobbying weights, or ratification probabilities.

A second group of papers provides the quantitative trade-policy tools used to discipline the counterfactuals. [Caliendo and Parro \(2010\)](#) study Mercosur partner choice and show why intermediate-input access matters for welfare comparisons inside a trade bloc. [Costinot and Rodríguez-Clare \(2014\)](#) explain how quantitative trade models discipline counterfactual welfare analysis. The model environment used here follows [Caliendo and Parro \(2015\)](#), which provides a multi-country, multi-sector framework with input-output linkages. [Ossa \(2016\)](#) reviews quantitative commercial-policy models and their use in tariff counterfactuals, while [Bagwell et al. \(2021\)](#) combine quantitative trade methods with multiparty tariff negotiations. In the Mercosur context, [Campos and Timini \(2022\)](#) show that gains can differ sharply across members. I differ by using the final signed EU–Mercosur tariff schedule as a fixed counterfactual object and asking how its welfare and sectoral effects change across benchmark years and

external scenarios.

A third literature thread helps explain why the final tariff schedule may be interpreted differently in a world shaped by China exposure, U.S. tariff risk, and strategic-input concerns. [Autor et al. \(2013\)](#) show how import competition can generate concentrated local adjustment costs. For Brazil, [Costa et al. \(2016\)](#) study winners and losers from the China trade boom, while [Paz \(2018\)](#) and [Mesquita Moreira et al. \(2020\)](#) study manufacturing responses to Chinese import competition. This evidence motivates the paper’s attention to sectoral exposure rather than aggregate welfare alone. The external-risk scenario connects to evidence on supply-chain reallocation after geopolitical shocks ([Alfaro and Chor, 2023](#)) and on heterogeneous effects of U.S. tariff shocks across firms, consumers, and sectors ([Fajgelbaum et al., 2020](#); [Amiti et al., 2020](#)). [Santacreu et al. \(2023\)](#) emphasizes how exposure shapes tariff-policy effects, and [Cavalcanti et al. \(2025\)](#) studies diverted demand toward Brazilian regions. The rare-earth exercise is narrower: it uses production-network logic and rare-earth supply-chain risk to compare input-cost exposure with the direct tariff component.

2 Institutional Framework

2.1 History, Barriers, and Gains

By the time the agreement entered 2026, it carried a long institutional history that had become part of the economic question. The framework cooperation agreement entered into force in 1999, trade negotiations began in 2000, and talks were relaunched in May 2016 after several years of limited progress ([European Commission, 2026d](#)). The trade pillar reached a first political agreement on 29 June 2019, and the political and cooperation components were concluded in 2020 ([European Commission, 2026b](#)). The fact that this text still did not move directly to ratification matters for the exercise. Environmental enforcement, deforestation, agricultural market access, and the legal form of the agreement remained contested, so the agreement continued to be evaluated in a changing economic and political environment ([Grieger, 2020](#)).

The final negotiation phase changed the institutional setting, but it did not remove the political constraints surrounding the agreement. The parties reopened parts of the text in 2023–2024 and announced an improved political agreement in Montevideo on 6 December 2024 ([MERCOSUR, 2024](#)). Brazil’s official summary describes new sustainable-development provisions, procurement adjustments, and longer automotive phase-outs, including special treatment for electric, hydrogen, and new-technology vehicles ([Government of Brazil, 2024](#)). The final text was separated into two instruments: the EU–Mercosur Partnership Agreement (EMPA) and the interim Trade Agreement (iTA). The Council authorised signature on 9 January 2026, and both instruments were signed on 17 January 2026 ([Council of the European Union, 2026b](#)). EU customs guidance records provisional application of the iTA on 1 May 2026 after the required legal steps ([European Commission, Directorate-General for Taxation and Customs Union, 2026](#)). These steps turned a long-negotiated text into an implementable tariff schedule, but they did not eliminate the sectoral concerns that shaped the debate.

Seen through the model, these political barriers map into sectoral interests rather than a single aggregate welfare concern. In the European Union, opposition centered on agriculture, especially beef, poultry, sugar, ethanol, rice, honey, maize, and related products. The Amazon fires in August 2019 linked market access to climate and deforestation concerns ([Al Jazeera and News Agencies, 2019](#); [Grieger, 2020](#)). Farmer mobilization continued after the 2024 agreement, including demonstrations in France in November 2024 ([Associated Press, 2024](#)) and a tractor protest in Paris in January 2026 ([Euronews, 2026](#)). The Council later adopted a separate agricultural safeguard regulation in March 2026 ([Council of the European Union, 2026a](#)). In Mercosur, the main documented concerns were reciprocal liberalization in manufacturing and the preservation of policy space in areas such as procurement and automobiles ([Government of Brazil, 2024](#)). These facts motivate the choice to report sectoral losses and exposure alongside aggregate welfare.

2.2 How the Deal Works

The tariff counterfactual uses Annex 2-A because it is the part of the final agreement that can be mapped most directly from legal commitments into the model. It lists product-level tariff commitments rather than a uniform full liberalization rule ([European Union and Mercosur, 2026a](#)). Official summaries describe the agreement as eliminating tariffs on most products exported by the European Union to Mercosur, including current Mercosur duties of up to 35 percent on cars, 14–20 percent on machinery, and up to 14 percent on pharmaceuticals ([European Commission, 2026c](#)). The legal schedule also combines immediate eliminations, staged reductions, exclusions, tariff-rate quotas, safeguards, and special treatments. For that reason, the benchmark uses only the Annex 2-A tariff schedule, while the zero-tariff case is a separate full-liberalization upper-bound exercise rather than a historical alternative version of the agreement.

Exclusions, quotas, and safeguards are part of the agreement’s distributional design because they limit liberalization in politically exposed sectors while also reducing the market-access gains available to the other side. On EU imports from Mercosur, protected products are mainly agricultural and food products such as beef, poultry, sugar, ethanol, rice, honey, dairy, eggs, garlic, starches, and biodiesel. On Mercosur imports from the European Union, sensitive margins include dairy, garlic, automobiles, and selected manufacturing lines. EU agricultural safeguards lower Mercosur’s potential agricultural gains, while Mercosur protections in autos and selected manufacturing reduce potential EU industrial gains. Appendix 7 reports the tariff-line counts, quota magnitudes, excluded sectors, and rare-earth-related lines used to code the tariff shock.

Rare-earth-related products appear in the goods schedule, but they appear as ordinary tariff lines. The extracted Annex 2-A lines for HS6 253090, 280530, 284610, and 284690 receive standard tariff treatment in both directions. On Mercosur imports from the European Union, the legal base rates are about 4 percent for the broader mineral-substances line and 2–10 percent for rare-earth metals and compounds. On EU imports from Mercosur, HS6 253090 is already free, while the rare-earth metal and compound lines face positive base rates. These lines enter through the ordinary goods-tariff schedule.

Some provisions matter for interpretation even though they are not the tariff shock itself. Article 2.9 restricts export duties after the transition period, while Articles 2.7, 2.10, and 2.11 discipline export licensing, export monopolies, quantitative restrictions, and export-price requirements ([European Union and Mercosur, 2026c](#)). Annex 2-B gives EU-bound exports preferential treatment if Brazil uses export duties on HS chapters 25–28, including the rare-earth proxy lines ([European Union and Mercosur, 2026b](#)). The Commission’s critical-raw-materials factsheet presents these provisions as improving supply-chain security by reducing export-tax risk, minimum-pricing requirements, and export monopolies ([European Commission, 2026a](#)). I use these provisions to motivate the rare-earth scenario, while treating its magnitude as a maintained input-cost assumption.

To translate the rare-earth concern into the model, the relevant economic channel is input-cost exposure. Critical-raw-material policy concerns are documented by the International Energy Agency and the European Commission ([International Energy Agency, 2026](#); [European Commission, 2026e](#)). Upstream input-cost changes can propagate to downstream sectors through input-output linkages ([Baqae and Farhi, 2022](#); [Alfaro et al., 2025](#)). Appendix 7 reports the legal lines, and Appendix 11 reports the rare-earth access and risk scenarios. Keeping these pieces separate is important because the legal schedule identifies tariff changes, while the rare-earth exercise measures exposure to a possible input-cost channel.

Other provisions shape market access and political interpretation even though they are not imposed as direct tariff shocks. Rules of origin define which goods qualify for preferences through originating-good criteria, cumulation, tolerances, product-specific rules, statements on origin, recordkeeping, and verification. Government procurement opens covered tenders subject to listed entities and thresholds; the Commission highlights Brazil’s federal procurement market, which exceeds EUR 8 billion per year ([European Commission, 2026c](#)). SPS and technical-barrier provisions keep the importing party’s standards in force, including establishment approval, import checks, audits, and emergency measures. The sustainable-development text covers Paris Agreement

implementation, labour standards, biodiversity, forests, and deforestation. These provisions are central to the agreement, but the quantitative exercise uses the tariff schedule because it is the component that can be coded consistently across products, sectors, and benchmark years.

The legal-text review first identifies the relevant Annex 2-A tables and preserves staging labels, exclusions, tariff-rate quotas, safeguards, and special treatments. The row-level extraction is audited: the extractor reads the official PDF, uses page coordinates to recover table columns, and records source pages for audit. Second, the extracted CN and NCM legal rows are normalized to HS6 products and assigned final cut shares, with non-simple legal treatments coded conservatively unless a simple ad valorem cut can be recovered. Third, each HS6 cut share is applied to its MACMAP baseline tariff. Fourth, the post-agreement tariffs are aggregated to the 49-sector model using BACI trade weights before solving the counterfactual. Appendix 7 gives the detailed audit trail, concordance checks, and missing-mapping diagnostics.

3 Theoretical Framework

A preferential tariff cut does not have the same value in every benchmark economy. Its effect depends on who trades with whom, which sectors are exposed, how production uses intermediate inputs, and what outside options are available when the agreement is implemented. This is the link between the long delay and the model. The EU–Mercosur agreement can become easier to defend even if the legal tariff schedule is unchanged, because the economic environment around that schedule has changed.

3.1 Mechanism and Counterfactual Object

The quantitative exercise starts from the proposed tariff cuts. The goods schedule changes tariffs between the European Union and the modeled Mercosur economies, Argentina and Brazil. It can be mapped from legal text to product lines, from product lines to sectors, and from sectors to a quantitative trade model. The paper therefore measures the signed goods-tariff component as a counterfactual shock in each benchmark economy. The separate zero-tariff exercise removes bilateral goods tariffs completely in the same benchmark years and serves as a full-liberalization comparison. Other provisions on tariffs remain important for interpretation, but they are not imposed as direct model shocks.

The result of the model may present a small aggregate gain from the tariff change, but this does not make the agreement irrelevant; it shifts the analysis toward distribution, external risk, and strategic-input exposure. If direct welfare gains are small, the economic case depends on whether the previous sectoral resistance margins weaken, whether the agreement becomes more useful when outside markets become less reliable, and whether strategic-input exposure becomes large enough to matter. These are the margins that connect the tariff schedule to the timing of the agreement.

The tariff mechanism can be read as a chain from delivered costs to sourcing, prices, production costs, and welfare. Annex 2-A changes delivered costs for affected goods, and those cost changes alter sourcing shares and sectoral prices. Sectoral price changes then move through production costs because sectors use each other’s outputs as inputs. The final effects appear in real income, sectoral output, import reallocation, and the concentration of losses. Because trade shares, sector sizes, initial tariffs, input-output links, and outside suppliers differ across years, the final signed schedule can produce different economic outcomes across benchmark economies.

The ambiguity is expected in customs-union theory, where preferential agreements can create welfare gains through trade creation and losses through trade diversion. [Viner \(1982\)](#) provides the classic distinction between trade creation and trade diversion. [Meade \(1955\)](#) develops the welfare logic of customs unions, while [Lipsey \(1957\)](#) clarifies why the welfare effect depends on the structure of consumption and substitution, not only on the presence of preferential tariff cuts. Political-economy models add that concentrated sectoral losses can matter even when

aggregate effects are positive (Krishna, 1998; Facchini et al., 2021). I use this logic to interpret loss concentration, not to estimate lobbying or votes.

To connect the tariff benchmark with the changed global environment, the framework adds two external margins. A more protectionist U.S. environment can raise the value of EU–Mercosur access for selected sectors, even if aggregate welfare barely changes. Rare-earth minerals can matter through input costs rather than through ordinary tariff lines. Both cases are scenario exercises around the Annex 2-A tariff benchmark. They are useful because they show how the agreement may become more defensible when the surrounding environment becomes more uncertain.

3.2 Model Environment and Main Equations

To trace those effects jointly, I use the multi-country, multi-sector, input-output Ricardian model of Caliendo and Parro (2015). This model captures value-chain adjustments that matter for the rare-earth and U.S. tariff-risk exercises. The same tariff change can alter sourcing, sectoral prices, and production costs at the same time, so the model provides a standard environment for tracing those effects together.

Countries are indexed by n and i , sectors by j , HS6 products by k , legal tariff lines by ℓ , and benchmark years by t . Importer n buys sector j goods from exporter i . In scenario notation, B denotes the observed benchmark economy and A denotes only the coded Annex 2-A goods-tariff counterfactual, not the full agreement. Let $\pi_{ni,t}^j$ denote the baseline expenditure share, $c_{i,t}^j$ the unit cost of production, $\delta_{ni,t}^j$ iceberg trade costs, $\tau_{ni,t}^j$ the ad valorem tariff, $d_{ni,t}^j = (1 + \tau_{ni,t}^j)\delta_{ni,t}^j$ total trade costs, and $\theta_{j,t}$ the trade elasticity. The baseline trade share is

$$\pi_{ni,t}^j = \frac{T_{i,t}^j \left(c_{i,t}^j d_{ni,t}^j \right)^{-\theta_{j,t}}}{\sum_h T_{h,t}^j \left(c_{h,t}^j d_{nh,t}^j \right)^{-\theta_{j,t}}}, \quad (1)$$

where $T_{i,t}^j$ summarizes sectoral productivity. These shares summarize the initial exposure to EU, Chinese, U.S., and rest-of-world suppliers. A tariff cut matters more when it affects a relationship that is already important in baseline sourcing or connected to important downstream sectors.

In exact-hat form, the counterfactual sectoral price index, trade share, and unit cost changes satisfy

$$\hat{P}_{n,t}^j = \left[\sum_i \pi_{ni,t}^{j,B} \left(\hat{c}_{i,t}^j \hat{d}_{ni,t}^j \right)^{-\theta_{j,t}} \right]^{-1/\theta_{j,t}}, \quad (2)$$

$$\frac{\pi_{ni,t}^{j,A}}{\pi_{ni,t}^{j,B}} = \left(\frac{\hat{c}_{i,t}^j \hat{d}_{ni,t}^j}{\hat{P}_{n,t}^j} \right)^{-\theta_{j,t}}, \quad (3)$$

$$\hat{c}_{i,t}^j = \hat{w}_{i,t}^{\alpha_{i,t}^j} \prod_k \left(\hat{P}_{i,t}^k \right)^{\gamma_{i,t}^{kj}}. \quad (4)$$

Here $\alpha_{i,t}^j$ is the value-added share and $\gamma_{i,t}^{kj}$ is the input share of sector k in sector j . Equations (2)–(4) connect tariff changes to prices, sourcing, and production costs. They are what makes the exercise general equilibrium rather than simple tariff accounting.

In the main Annex 2-A exercise, the agreement enters only through tariffs:

$$\hat{d}_{ni,t}^j(A) = \frac{1 + \tau_{ni,t}^{j,A}}{1 + \tau_{ni,t}^{j,B}}, \quad \tau_{ni,t}^{j,A} = \sum_{k \in \mathcal{H}(j)} w_{ni,k,t} \tau_{ni,k,t}^B (1 - c_{ni,k}), \quad (5)$$

where $c_{ni,k}$ is the legal HS6 cut share and the weights sum to one within each sector. Outside the EU–Argentina and EU–Brazil agreement margins, tariffs remain unchanged. Iceberg costs are fixed in the benchmark. This restriction is deliberate: it isolates the goods-tariff schedule before comparing that component with broader external and strategic-input scenarios.

Aggregate welfare is real income relative to the benchmark:

$$\hat{W}_{n,t}(A) = \frac{\hat{I}_{n,t}(A)}{\prod_j \left(\hat{P}_{n,t}^j(A) \right)^{\beta_{n,t}^j}}, \quad \Delta W_{n,t}(A) = 100 [\hat{W}_{n,t}(A) - 1], \quad (6)$$

where $\beta_{n,t}^j$ are final-expenditure shares. This is the aggregate tariff-component result. It is necessary, but it is not enough: small welfare changes can coexist with concentrated sectoral losses.

In Argentina and Brazil, the resistance margin I want to measure is legacy manufacturing. Let $\Delta q_{n,j,t}(A) = 100[\hat{Q}_{n,j,t}(A) - 1]$ be the sectoral output change, let $s_{n,j,t}$ be the sector's baseline output share, and let $\mathcal{J}^{mfg}$ denote the legacy manufacturing bundle. The loss mass and loss concentration are

$$R_{n,t}^{mfg}(A) = \sum_{j \in \mathcal{J}^{mfg}} s_{n,j,t} \max\{0, -\Delta q_{n,j,t}(A)\}, \quad (7)$$

$$S_{n,t}^{mfg}(A) = \frac{R_{n,t}^{mfg}(A)}{\sum_j s_{n,j,t} \max\{0, -\Delta q_{n,j,t}(A)\}}, \quad (8)$$

with $S_{n,t}^{mfg}(A)$ undefined if total sectoral losses are zero. These two measures say whether losses in exposed manufacturing sectors are large and whether they account for a large share of all negative sectoral effects. For the European Union, I report bloc-level agri-food output effects plus a member-state agri-food exposure diagnostic, because EU member states are not separate regions in the structural model.

Rare-earth minerals enter the model through input costs because the concern in this part is supply-chain exposure rather than an additional tariff provision. The scenario modifies unit costs as

$$\hat{c}_{i,t}^{j,REE} = \hat{c}_{i,t}^j \left(\hat{P}_{i,t}^{REE} \right)^{\rho_{i,j,t}^{REE}}, \quad (9)$$

where $\hat{P}_{i,t}^{REE}$ is the rare-earth input-price change and $\rho_{i,j,t}^{REE}$ is sector j 's direct exposure to the upstream mining and chemicals sectors that contain the rare-earth HS6 bundle. In the preferred interpretation, $\hat{P}_{EU,t}^{REE} < 1$ represents lower expected delivered costs for rare-earth inputs when export-tax, export-licensing, quota, monopoly, or minimum-price risks are partly insured. The production-network logic follows [Baqae and Farhi \(2022\)](#), and the rare-earth export-restriction mechanism follows [Alfaro et al. \(2025\)](#). Appendix 11 defines $\hat{P}_{i,t}^{REE}$ and $\rho_{i,j,t}^{REE}$ and reports the diagnostic cases.

To check whether the location of the primitive drives the welfare pattern, the Argentina–Brazil placement moves the same idea upstream in Mercosur. Instead of lowering the EU input price directly, it lowers the delivered cost of Argentina and Brazil's B08 and C20 output by $(1 + g)^{-\omega_{j,t}}$, where g is the maintained production-cost change and $\omega_{j,t}$ is the rare-earth bundle weight of sector j . Comparing the two placements shows how much the welfare pattern depends on where the strategic-input primitive is located.

3.3 Measured Economic Margins

For the empirical section, the timing question becomes four margins: the size of the direct tariff gain, the location of sectoral losses, the value of partner access when U.S. tariff risk rises, and the scale of rare-earth input-cost

exposure.

1. **The first measured margin is tariff-component welfare, which gives the direct welfare value of the coded goods-tariff schedule change.** The aggregate welfare effect of Annex 2-A is $\Delta W_{n,t}(A)$. I also report the trade-weighted mean sector effective tariff cut on positive-value agreement-pair trade. The Annex 2-A, internal-tariff, and zero-tariff cases show how much can be explained by tariff-only changes, with the zero-tariff case serving as the full-liberalization benchmark.
2. **The second measured margin is the distribution of sectoral gains and losses, especially the resistance margins in Argentina-Brazil manufacturing and EU agri-food exposure.** For Argentina and Brazil, the focus is the output loss in legacy manufacturing sectors and the concentration of those losses. The margin weakens when the loss mass falls,

$$R_{n,t_2}^{mfg}(A) < R_{n,t_1}^{mfg}(A),$$

or when losses become less concentrated,

$$S_{n,t_2}^{mfg}(A) < S_{n,t_1}^{mfg}(A),$$

for a later benchmark year t_2 relative to an earlier benchmark year t_1 . For the European Union, the corresponding margin is the agri-food effect at the bloc level and the member-state exposure diagnostic.

3. **The third measured margin links external reallocation to U.S. tariff risk through baseline trade shares, sector sizes, and sourcing options.** China and the United States affect the counterfactual through baseline trade shares, sector sizes, and sourcing options. The U.S. tariff-risk scenario puts this logic in a 2022 policy environment with higher U.S. goods-import tariffs and asks whether Annex 2-A partly offsets sectoral output losses.
4. **The fourth measured margin compares the tariff schedule with rare-earth input-cost exposure.** The fourth margin is the additional welfare and sectoral output effect from rare-earth input-cost exposure. The EU input-cost placement compares

$$\Delta W_{n,t}(A, R^{EU}) - \Delta W_{n,t}(A),$$

while the Argentina–Brazil production-cost placement compares

$$\Delta W_{n,t}(A, R^{AB}) - \Delta W_{n,t}(A).$$

The comparison shows whether strategic-input exposure can be economically larger than the direct tariff component.

Taken together, these margins separate four questions that are often blurred in the policy debate: how large the tariff effect is, where losses are concentrated, whether the agreement helps under U.S. tariff risk, and whether rare-earth input-cost exposure changes the value of the agreement.

3.4 Counterfactual Exercise, Calibration, and Interpretation

For each benchmark year, the model is solved at the observed baseline and after the Annex 2-A tariff change. The policy shock follows (5). It applies only to bilateral goods trade between the European Union and Argentina and between the European Union and Brazil. All other tariffs and iceberg costs remain at their benchmark values. The reported solves use the same zero-aggregate-deficit closure in the baseline and counterfactual equilibria.

The timing exercise uses the final legal schedule as a common counterfactual shock while recalibrating the benchmark economy in 2007, 2011, 2014, 2017, and 2022. The legal schedule is fixed, but observed trade shares,

tariffs, input-output linkages, sector sizes, and elasticities vary by year. This is the core link between the long delay and the quantitative exercise: the agreement is the same, but the economy around it is not.

To ask whether the agreement is more valuable in a more protectionist outside environment, the U.S. tariff-risk scenario changes the 2022 policy setting by adding 10 percentage points to U.S. goods-import tariffs against non-U.S. origins, motivated by the 2025 U.S. reciprocal-tariff policy environment ([Federal Register, 2025](#); [The White House, 2025b](#)). The incremental value of Annex 2-A in that environment is

$$\Delta X_{n,2022}^{US}(A) = X_{n,2022}(A; U) - X_{n,2022}(B; U), \quad X \in \{\Delta W, \Delta q\}.$$

This comparison shows whether the agreement has a larger sectoral effect when a major outside partner becomes more protectionist. The high-intensity 25 percentage-point case remains a robustness case in the appendix.

The rare-earth minerals exposure scenario keeps the tariff schedule and the Caliendo–Parro solve fixed, then adds the upstream input-cost term in (9) or the production-cost placement defined in Appendix 11. The 10 percent EU input-cost change is a scenario parameter for avoided export-tax or export-restriction price premia, not an estimated legal coefficient. The Argentina–Brazil placement asks how the pattern changes when the primitive is modeled as lower upstream production costs in B08 and C20. A network-exposure diagnostic reports the direct-plus-indirect input-output footprint of the same primitive, but the counterfactual uses the direct exposure term to avoid double-counting network propagation.

After these comparisons, the interpretation becomes: welfare is the real-income object in (6), sectoral resistance summarizes the loss concentration measured by (7) and (8), and partner reallocation shows how the fixed schedule operates in economies with different outside suppliers. The U.S. tariff-risk and rare-earth scenarios evaluate external and strategic-input channels around the same legal agreement. The model does not explain ratification. It shows how the agreement can become economically more defensible even when the direct tariff component remains small.

4 Data and Calibration

The data construction keeps two objects separate: the benchmark economy observed in each calibration year and the tariff shock implied by the final agreement text. The benchmark economy comes from production links, bilateral trade shares, applied tariffs, sector sizes, and trade elasticities. Annex 2-A then supplies the legal cut shares that are mapped into that economy. Appendix 8 reports the sector aggregation, elasticity tables, and benchmark-object checks behind this section.

4.1 Data Inputs

The benchmark economy is built from sources that each fix a different part of the model. OECD ICIO provides production, intermediate input use, final demand, value added, and bilateral trade values, which determine the production network and the size of each sector ([Organisation for Economic Co-operation and Development, 2023](#)). BACI supplies HS6 product-level trade values, so it is used to weight tariff lines and to decide how detailed products enter the model sectors ([Gaulier and Zignago, 2010](#)). MACMAP supplies applied HS6 tariffs for the baseline tariff matrix ([Guimbard et al., 2012](#)).¹ WITS/UNCTAD TRAINS enters only in the elasticity calibration, where the tariff panel is cleaned to the same HS6 country-pair structure and combined with BACI trade weights ([World Bank and UNCTAD, 2026](#)).

The five benchmark years, 2007, 2011, 2014, 2017, and 2022, are the years in which the required production, trade, and tariff objects can be assembled consistently. They also span earlier and later benchmark environments, which

¹For the final benchmark year, the build uses exact 2022 MACMAP files when available and 2023 observations as proxies in two Brazil files. Product-level source years are not fully retained in the final imputed tariff surface.

is what makes the timing comparison possible. The regional aggregation is Argentina, Brazil, China, the European Union, the United States, and the rest of the world, with fixed EU-27 membership in all years. The model is solved in 49 sectors, while the 8-sector aggregation is used only to report results and organize appendix tables.

Table 1: Core Data Sources and Benchmark Coverage

Source	Raw unit / resolution	Role in the model
OECD ICIO	Country-sector IO	Production network, final demand, value added, and bilateral trade
BACI	HS6 goods trade	Product weights for tariff aggregation
MACMAP	HS6 tariffs	Applied tariff matrix before completion and sector aggregation
WITS/TRAINS	Country-partner HS6 tariffs	Country-level tariff variation used for the supported elasticity updates

Notes: The table reports the raw inputs used in the main counterfactuals. Benchmark years, region definitions, and sector aggregation are described in the text and appendix.

4.2 Tariffs and Agreement Mapping

The tariff data link product-level legal commitments to the sectoral model, which requires preserving tariff-line information while producing sector-level shocks. I therefore keep tariffs at HS6 until the final aggregation step, using MACMAP for applied rates and BACI trade values as the weights that move products into model sectors (Gaulier and Zignago, 2010; Guimbard et al., 2012). The missing-tariff rule follows the measurement concern emphasized by Teti (2024): missing tariff cells should not be treated mechanically as zero when local product or importer information is available. Let $\mathcal{H}(j)$ denote the HS6 products mapped to model sector j . The baseline tariff entering the model is

$$\tau_{ni,t}^{j,B} = \sum_{k \in \mathcal{H}(j)} w_{ni,k,t} \tau_{ni,k,t}^B,$$

where $\tau_{ni,k,t}^B$ is the completed MACMAP tariff and $w_{ni,k,t}$ is the BACI trade weight for importer n , exporter i , product k , and year t . Missing tariff cells are completed before aggregation using the documented hierarchy in Appendix 8; zero is used only as the residual fallback.

After the product-level benchmark is built, Annex 2-A supplies the legal cut shares used to construct the counterfactual tariff (European Commission, 2026b). I convert the legal CN and NCM tariff-line schedules into HS6 cut shares, apply each cut to its baseline tariff, then aggregate the post-agreement tariffs to the 49-sector model using the same BACI weights. The counterfactual tariff is

$$\tau_{ni,t}^{j,A} = \sum_{k \in \mathcal{H}(j)} w_{ni,k,t} \tau_{ni,k,t}^B (1 - c_{ni,k}),$$

where $c_{ni,k}$ is the HS6 legal cut share. The sector effective cut is $C_{ni,t}^j = 1 - \tau_{ni,t}^{j,A} / \tau_{ni,t}^{j,B}$ when the baseline tariff is positive, and zero otherwise. The shock applies only to bilateral goods tariffs between the European Union and Argentina and between the European Union and Brazil. Quotas, exclusions, safeguards, and other non-simple legal treatments are coded conservatively as described in Appendix 7. The zero-tariff exercise removes remaining bilateral goods tariffs and is used only as a robustness check.

4.3 Calibration

To calculate the elasticities, I start from the Caliendo–Parro sectoral elasticity vector mapped to the 49-sector model, since that vector gives every sector a disciplined quantitative-trade value before any country-level update is attempted (Caliendo and Parro, 2015). I then estimate updates from WITS/TRAINS country-level HS6 variation using a CP-style triple-ratio calculation. A WITS value is accepted only when the relevant country-level panel contains enough observations, HS6 products, country triples, and tariff variation to identify the elasticity at its declared level of aggregation.

A WITS estimate replaces the mapped value only when it passes the support checks; otherwise the sector keeps the original mapped value. The mapped value is retained when the support checks fail at every WITS level. Unsupported sectors are not dropped from the model, and weak tariff variation is not converted into a new elasticity. All 49 sectors receive an elasticity in every year; WITS can update only the HS6 goods sectors with enough country-level tariff variation.

The final benchmark therefore uses a five-step hierarchy, but the hierarchy is meant to discipline weak support rather than mechanically shrink accepted WITS estimates. The hierarchy first uses the strict WITS sector-year estimate, then a same-sector estimate pooled across all benchmark years, then a broader goods-group estimate by year, then a broader goods-group estimate pooled across years, and only then keeps the mapped Caliendo–Parro value. Appendix 8 reports the hierarchy and the literature basis for this rule.

Table 2: Calibration Summary by Benchmark Year

Year	Sectors	WITS based	CP value	θ_{min}	θ_{med}	θ_{mean}	θ_{max}
2007	49	15	34	0.43	8.22	7.78	23.72
2011	49	12	37	0.43	8.22	8.78	64.85
2014	49	13	36	0.22	8.22	8.28	64.85
2017	49	15	34	0.26	8.22	8.38	64.85
2022	49	20	29	0.41	8.22	7.82	64.85

Notes: ‘WITS based’ counts sectors whose elasticity is a supported WITS country-level CP estimate at one of the hierarchy levels. ‘CP value’ counts sectors kept at the mapped Caliendo–Parro value. The reported moments are based on the final yearly elasticity vector used in each benchmark object.

At the end of the calibration, each benchmark year combines the ICIO production network, the HS6-based tariff matrix, the baseline trade and sector-size objects, and the WITS hierarchy elasticity vector. The calibrated six-region objects pass a no-shock equilibrium consistency check in all five benchmark years. The rare-earth and U.S. tariff exercises use the same calibrated objects and add only the scenario-specific input-cost or tariff changes described in Appendix 11 and Appendix 10.

5 Results

The results begin with the finding that the Annex 2-A is not a large aggregate welfare shock. That does not make the tariff exercise uninformative. It changes the question: if average welfare gains are small, the relevant evidence comes from where losses fall, how partner sourcing changes, and whether the 2022 external environment makes some margins more important than they were in earlier benchmark years. The benchmark result remains the Caliendo–Parro equilibrium effect of the coded goods-tariff schedule only. The U.S. tariff-risk and rare-earth exercises are kept separate, because they ask how the same tariff benchmark should be interpreted under maintained external-risk and input-cost scenarios.

5.1 Sectoral Resistance Margins in Annex 2-A

At the sector level, Annex 2-A looks less like a broad national-income shock and more like an uneven market-access exchange. Argentina and Brazil expand most visibly in extraction and mining support, but they contract in a recognizable manufacturing bundle: basic metals, non-ferrous metals, metal products, autos, electrical equipment, machinery, and related manufacturing. These are model output changes, not estimates of lobbying or ratification behavior. They identify the sectors where a tariff agreement with small aggregate effects can still create visible resistance. On the European side, the comparable sensitive object is agriculture and agri-food exposure, which has to be read with more care because the European Union enters the model as one region.

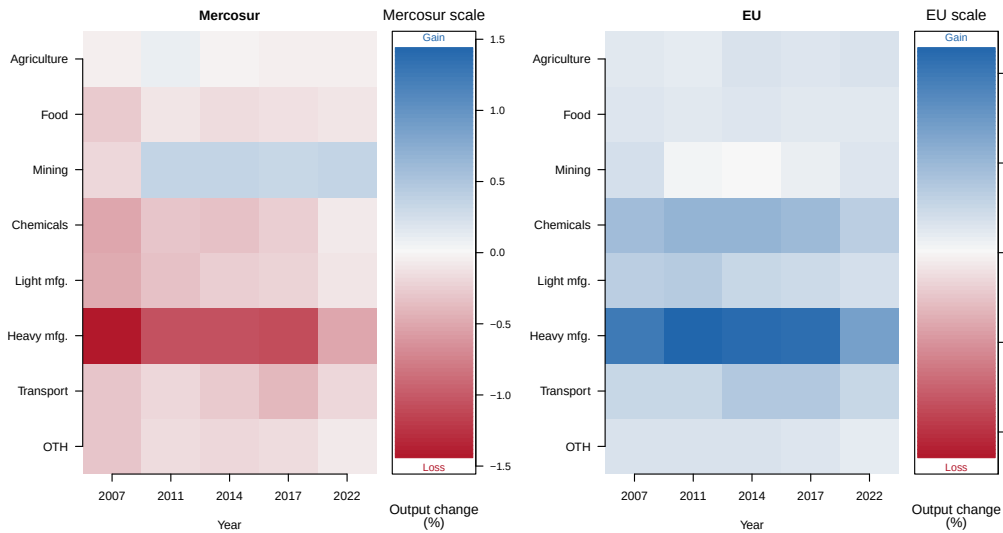


Figure 1: Eight-Sector Output Responses Under Annex 2-A

Notes: The 49 model sectors are aggregated to eight reporting groups. Mercosur sums Argentina and Brazil before computing the percent change. Red cells are output losses and blue cells are output gains. Each panel uses its own symmetric color scale centered at zero, so colors show relative sectoral output changes within the panel rather than a common Mercosur–EU magnitude.

Table 3: Largest Sectoral Output Responses Under Annex 2-A

Region	Year	Top gains	Top losses
Argentina	2007	B09 mining support (1.73); B06 extraction (0.98)	C24A metals (-4.18); B07 metal ores (-3.28); C24B non-ferrous metals (-2.87); C25 metal products (-1.55); C27 electrical equip. (-1.41)
Argentina	2022	B09 mining support (1.85); B06 extraction (1.77); C19 refined petroleum (0.13); H50 water transport (0.11); B05 coal (0.04)	B07 metal ores (-1.50); C24A metals (-1.18); C24B non-ferrous metals (-0.74); C16 wood products (-0.51); C25 metal products (-0.35)
Brazil	2007	B09 mining support (0.67); B06 extraction (0.23); H50 water transport (0.12)	B07 metal ores (-3.49); C24B non-ferrous metals (-3.11); C24A metals (-3.07); B05 coal (-1.55); C25 metal products (-1.34)
Brazil	2022	B06 extraction (1.71); B09 mining support (1.45); C19 refined petroleum (0.50); B05 coal (0.28); H50 water transport (0.18)	C24A metals (-1.52); B07 metal ores (-1.36); C24B non-ferrous metals (-0.91); C25 metal products (-0.33); C27 electrical equip. (-0.25)
European Union	2007	C24A metals (0.42); C24B non-ferrous metals (0.35); B07 metal ores (0.31); C27 electrical equip. (0.25); C25 metal products (0.17)	B09 mining support (-0.15); B06 extraction (-0.02); C301 ships (-0.01)
European Union	2022	B07 metal ores (0.59); C24A metals (0.42); C24B non-ferrous metals (0.27); B08 other mining (0.12); C25 metal products (0.11)	B09 mining support (-0.06); B06 extraction (-0.06); H53 postal/courier (0.00); H50 water transport (0.00); J58T60 media (0.00)

Notes: All entries are percent deviations from the same-year benchmark CP equilibrium unless explicitly labeled as scenario increments. Within each region-year, sectors are ranked by the size of the positive or negative output response. The table reports up to five positive gains and five negative losses for Argentina, Brazil, and the European Union in the endpoint years. Values rounded to zero can have either sign before rounding.

The structural model does not produce a large EU agriculture loss, even though agriculture remains politically visible in the agreement debate. Agriculture moves from 0.021 percent in 2007 to 0.031 percent in 2022, and food remains small and positive, from 0.025 to 0.021 percent. Because the European Union is an aggregated model region, I do not use the CP solve to rank member-state political losses. The narrow model claim is instead that

Annex 2-A does not generate a large EU bloc-level agriculture or food output loss, even though the sector remains politically visible in the agreement debate.

The member diagnostic explains why the political debate can look sharper than the bloc-level output result. Table 4 uses BACI imports from Argentina and Brazil in the same agriculture and food sectors, outside the structural solve. Germany and France have visible agri-food imports, at USD 4.5 billion and USD 1.7 billion in 2022, while the largest individual import-exposure rows are the Netherlands and Spain. Ireland and Romania are smaller in levels, but agri-food accounts for more than three quarters of their imports from the modeled Mercosur partners. The point is not that the model finds large EU agricultural losses or member-state output effects. It is that an aggregate EU region can hide member-state and product-level exposure that remains politically salient.

Table 4: Selected EU Member-State Agri-Food Exposure to Argentina and Brazil, 2022

Country or bloc	Agri-food imports	Own import share	EU agri-food share
Netherlands	5944	41.1	20.8
Spain	5284	45.4	18.5
Germany	4459	48.1	15.6
Italy	3257	46.6	11.4
France	1693	39.1	5.9
Poland	1534	53.2	5.4
Ireland	801	76.1	2.8
Romania	415	79.4	1.4
Rest of EU	5204	35.5	18.2

Notes: Agri-food imports are 2022 BACI imports from Argentina and Brazil in the model's agriculture and food sectors, reported in USD millions. Own import share is agri-food imports divided by all imports from Argentina and Brazil for the same country or bloc. EU agri-food share is the row's share of EU-27 agri-food imports from Argentina and Brazil. These are exposure diagnostics, not member-state CP output effects; the Netherlands row may partly reflect port and re-export activity.

5.2 Did the Argentina–Brazil Manufacturing Margin Weaken?

Whether the old Argentina–Brazil manufacturing concern weakened depends on what is being measured. The amount of manufacturing loss falls, but manufacturing remains a large share of total losses. Table 5 puts the aggregate gain next to the output-share-weighted manufacturing-loss mass. In Argentina, the welfare gain falls from 0.031 percent in 2007 to 0.009 percent in 2022, while the manufacturing loss mass falls from 0.136 to 0.035 percentage points. Manufacturing's share of all output-share-weighted loss mass falls slightly, from 28.8 to 27.1 percent. In Brazil, the loss mass falls from 0.069 to 0.032 percentage points, but the loss share rises, from 20.8 to 30.2 percent. The result is therefore a level reduction, not a disappearance of loss concentration. For the EU, the agri-food margin is not measured by an analogous member-state loss mass because the EU is one model region; it remains part of the political interpretation through the agriculture/food output diagnostic and the country-level agri-food exposure table above.

Table 5: Aggregate Gains and Manufacturing Loss Concentration Under Annex 2-A

Country	Year	Welfare gain (%)	Mfg. loss mass R^M	Mfg. loss share S^M (%)
Argentina	2007	0.031	0.136	28.8
Argentina	2022	0.009	0.035	27.1
Brazil	2007	0.024	0.069	20.8
Brazil	2022	0.016	0.032	30.2

Notes: Welfare gains and sectoral output changes use percent deviations from the same-year benchmark CP equilibrium unless explicitly labeled as scenario increments. The manufacturing-loss mass is $R_{n,t}^{mfg} = \sum_{j \in \mathcal{J}^{mfg}} s_{n,j,t} \max\{0, -\Delta q_{n,j,t}\}$, where $s_{n,j,t}$ is the sector's baseline output share and $\Delta q_{n,j,t}$ is the sectoral output change in percent. The share is $S_{n,t}^{mfg}$, the manufacturing-loss mass divided by total output-share-weighted loss mass across all contracting sectors.

This is why the agreement should not be summarized by aggregate welfare only. Positive endpoint welfare gains coexist with concentrated sectoral losses. From 2007 to 2022, the Argentina–Brazil manufacturing loss level becomes smaller, but the concentration of losses does not disappear and strengthens for Brazil. The EU side remains small in bloc-level output, but politically exposed through member-state and product-level agri-food imports.

5.3 What Reallocates Around the Resistance Margins?

The reallocation evidence shows the market-access side of those sectoral results. Argentina and Brazil gain most visibly in extraction and mining support, while their import sourcing moves toward the European Union. Annex 2-A raises Argentina and Brazil’s imports from the EU in both endpoint years and reduces imports from some outside partners. The China and U.S. columns report separate import responses, while the displacement share is only a compact China-plus-U.S. value-change proxy. In Argentina, that combined proxy equals 23.1 percent of the EU import gain in 2007 and 29.4 percent in 2022. In Brazil, the same ratio rises from 18.9 to 29.2 percent.

On the tariff-only margin, the European Union experiences a smaller sourcing shift than Argentina and Brazil. EU imports from Argentina and Brazil rise by 4.5 percent in 2007 and 2.6 percent in 2022, while EU imports from China, the United States, and the rest of the world move only slightly. The model therefore shows asymmetric partner reallocation: Argentina and Brazil receive a clearer EU sourcing reorientation, while EU import sourcing changes only modestly. Broader institutional channels and rare-earth input-cost exposure enter only through the separate exercises below and remain outside the Annex 2-A tariff estimand.

Table 6: Counterfactual Import Reallocation Under Annex 2-A

Importer	Year	EU/MER	CHN	USA	ROW	Displaced share
ARG	2007	24.1	-6.5	-3.3	-4.1	23.1
ARG	2022	7.6	-0.9	-1.4	-1.2	29.4
BRA	2007	19.9	-6.3	-3.2	-3.3	18.9
BRA	2022	9.0	-1.1	-1.7	-1.2	29.2
EU	2007	4.5	0.3	0.2	0.2	0.0
EU	2022	2.6	0.1	0.1	0.1	0.0

Notes: EU/MER is the agreement-partner import change: EU for Argentina and Brazil, and Mercosur, defined as Argentina plus Brazil, for the European Union. The EU/MER, CHN, USA, and ROW columns are percent changes in imports relative to the same-year benchmark. Displacement share is a value-change ratio: the China-plus-U.S. import value loss divided by the agreement-partner import value gain. It is a partner-reallocation proxy, not a full trade-creation/trade-diversion decomposition.

5.4 External Tariff Risk and Selective Sectoral Offsets

The U.S. tariff-risk exercise asks how the agreement is interpreted in a more protectionist outside environment. The U.S. tariff exercise is a uniform stress-test shock: it adds 10 percentage points to U.S. goods-import tariffs on all non-U.S. origins, following the scale in official 2025 U.S. reciprocal-tariff materials ([Federal Register, 2025](#); [The White House, 2025b](#)). It is not an implementation of the legal U.S. tariff schedule. It also does not decompose the difference between 2007 and 2022 into China exposure, tariffs, input-output structure, and sector size. It evaluates the 2022 economy under an additional external tariff increase and asks whether Annex 2-A offsets selected sectoral losses created by that outside shock. The U.S.-tariff-only shock lowers welfare levels for Argentina and Brazil, while Annex 2-A gives selected sectors an alternative trade margin with the EU.

At the aggregate level, the U.S. tariff-risk exercise leaves the welfare interpretation largely unchanged. The measured object is deliberately narrow: the incremental effect of Annex 2-A within each tariff environment. In the baseline 2022 world, Annex 2-A raises welfare by 0.0094 percentage points for Argentina and 0.0161 for Brazil. Under the U.S. +10pp tariff environment, the increments are 0.0090 and 0.0135. The exercise changes trade orientation and outside options, but it does not make the tariff component a large aggregate-welfare shock.

The U.S. tariff-risk exercise is more informative at the sector level, where the agreement offsets exposure selectively rather than providing broad aggregate insurance. The U.S. tariff environment does not cushion the Argentina–Brazil manufacturing loss margin. In the U.S.-tariff-only scenario, 35 Argentina sectors and 30 Brazil sectors lose output; adding Annex 2-A offsets none of these losses and worsens losses in all 35 and 30 sectors, respectively. Using the unweighted sum of sectoral percentage losses, there is no recovery in Argentina or Brazil. For the EU, it offsets 44 of 45 losing sectors and recovers about 15.1 percent of the corresponding loss sum. Figure 2 then shows the selected sectors where the U.S. tariff shock lowers output and Annex 2-A partially offsets that loss. For Argentina and Brazil, agriculture instead loses more output when Annex 2-A is added. Argentine agriculture moves from a 0.281 percent output loss under the U.S. tariff shock alone to a 0.325 percent loss when Annex 2-A is also active; Brazilian agriculture moves from -0.283 to -0.336 percent. For the European Union, the largest offset sectors are metal ores, iron and steel, non-ferrous metals, other mining, chemicals, and transport equipment. In Argentina and Brazil’s metals and manufacturing sectors, the U.S. tariff shock and EU liberalization often add pressure rather than offsetting each other.

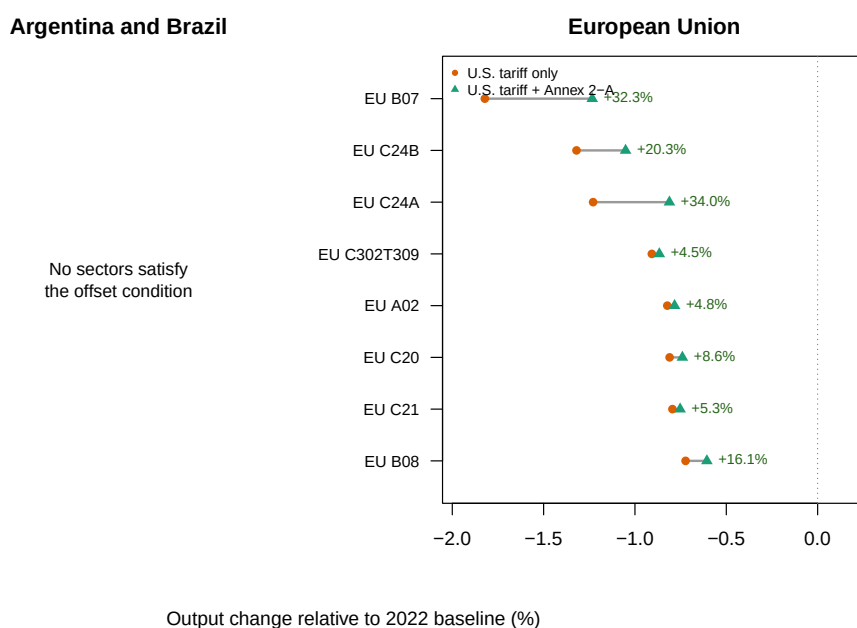


Figure 2: Sectors Where Annex 2-A Offsets U.S. Tariff Losses, 2022

Notes: The figure reports sectors in which the U.S. +10pp stress-test shock lowers output and adding Annex 2-A raises output relative to the U.S.-tariff-only scenario. No Argentina or Brazil sectors satisfy this condition, as indicated in the left panel; the right panel reports the eight EU offset sectors with the largest U.S.-tariff-only output losses. Values are sector output changes relative to the observed 2022 baseline. Offset shares are the fraction of the U.S.-tariff output loss recovered by adding Annex 2-A. The figure is selected on offset sectors and should be read with the text’s count of all U.S.-tariff loss sectors.

Table 7: Annex 2-A Under a U.S. +10pp Goods-Tariff Environment, 2022

Panel A. Welfare levels by scenario

Scenario	Argentina	Brazil	EU	USA
Annex 2-A only	0.009	0.016	0.005	0.000
U.S. global goods tariff +10pp	-0.030	-0.035	-0.038	0.128
Annex 2-A + U.S. global goods tariff +10pp	-0.021	-0.021	-0.033	0.127

Panel B. Incremental welfare effect of Annex 2-A

Region	Baseline world	U.S. +10pp world	Shift
ARG	0.0094	0.0090	-0.0003
BRA	0.0161	0.0135	-0.0025
EU	0.0048	0.0048	0.0000
USA	-0.0003	-0.0001	0.0001

Notes: All entries are percent deviations from the same-year benchmark CP equilibrium unless explicitly labeled as scenario increments. Panel A reports welfare levels relative to the observed 2022 benchmark for three scenarios: Annex 2-A only, U.S.-tariff-only, and Annex 2-A plus the U.S. tariff stress test. Panel B reports incremental Annex 2-A effects by subtracting the relevant no-agreement scenario from the agreement scenario in the same tariff environment.

5.5 Rare-Earth Minerals Exposure Exercise

Rare-earth minerals matter here because they point to an input-cost exposure channel that is not captured by tariff cuts alone. The exercise is not estimated from the agreement text and is not included in the Annex 2-A tariff estimand. It asks how large a maintained upstream-cost channel would be relative to the small legal tariff component. The legal motivation is export-side policy risk, not a hidden rare-earth tariff schedule: the goods chapter restricts export duties, export licensing, export monopolies, quantitative restrictions, and export-price requirements, while Annex 2-B gives EU-bound exports preferential treatment if Brazil uses export duties on HS chapters 25–28 and selected metal headings ([European Union and Mercosur, 2026c,b](#); [European Commission, 2026a](#)). This interpretation follows the China rare-earth dispute, where export taxes, quotas, and related restrictions raised trade-policy concerns ([World Trade Organization, 2014](#)). [Baqaee and Farhi \(2022\)](#) provides the production-network logic for upstream shocks, and [Alfaro et al. \(2025\)](#) connects rare-earth export restrictions to input-output linkages. The rare-earth proxy remains deliberately narrow, but its scale increases across benchmark years as strategic-input exposure becomes more important in the data. The related trade bundle rises from about USD 2.6 billion in 2007 to USD 14.4 billion in 2022. IEA evidence emphasizes China’s role in refining and downstream stages ([International Energy Agency, 2026](#)). USGS and European Commission sources support the critical-minerals classification used for the policy interpretation ([U.S. Geological Survey, 2025](#); [European Commission, 2026e](#)). The bundle should be read as a proxy, especially because HS6 253090 is a broad ore category rather than a clean rare-earth product. The sectoral bridge is direct input exposure in the EU production network: in 2022, exposure is highest in chemicals at 7.60 percent, rubber and plastics at 5.97 percent, other non-metallic minerals at 4.60 percent, other mining at 3.50 percent, and agriculture at 1.75 percent. This is the full role of the exercise in the main text: it scales export-risk-related input-cost exposure against the tariff benchmark, rather than treating rare earths as a separate Annex 2-A legal chapter or as evidence about the agreement’s signature.

The rare-earth results report two placements of the same input-cost idea: lower EU input costs and lower Argentina-Brazil upstream production costs. The illustrative 10 percent EU access scenario lowers the EU rare-earth input price index by 10 percent while Argentina and Brazil keep baseline access. This should be read as a maintained reduction in the expected delivered price of rare-earth-related inputs under lower export-tax, export-licensing, quota, monopoly, or minimum-price risk. Annex 2-A alone raises EU welfare by 0.005 percent in 2022; Annex 2-A plus the 10 percent access scenario raises it by 0.105 percent, an increment of 0.100 percentage points. The scaling result comes from Table 8’s 5, 10, and 20 percent sensitivity, not from treating the 10 percent price change as estimated. The alternative production-shock scenario lowers the effective delivered cost of Argentina and Brazil’s

B08 and C20 output instead. That shock mainly benefits the producers: under the 10 percent case, welfare rises relative to Annex 2-A alone by 0.208 percentage points in Argentina and 0.182 in Brazil, but only 0.001 in the EU. The interpretation is therefore simple: the access scenario is an EU rare-earth input-cost scenario, while the production shock is a Mercosur upstream-supply scenario. Both are outside the Annex 2-A tariff estimand and are used only to show how the country pattern of welfare effects changes when the input-cost primitive is placed in different parts of the production network.

Table 8: Rare-Earth Minerals Exposure Exercise: Two Input-Cost Placements

Panel A. Illustrative 10 percent primitive-to-outcome map

Primitive/object	Illustrative scaling case
REE bundle definition	HS6 253090, 280530, 284610, and 284690
Bundle source	BACI HS92 product descriptions in Gaulier and Zignago (2010) ; HS6 253090 is a broad mineral proxy that includes rare-earth ores below HS6. Strategic context: International Energy Agency (2026) , U.S. Geological Survey (2025) , and European Commission (2026e)
Sector mapping	HS6 253090 → B08 (Other mining and quarrying); HS6 280530, 284610, and 284690 → C20 (Chemicals and chemical products)
Illustrative EU input-price change	-10%
Top exposed EU sectors	C20 (Chemicals and chemical products, 7.60%); C22 (Rubber and plastic products, 5.97%); C23 (Other non-metallic mineral products, 4.60%); B08 (Other mining and quarrying, 3.50%); A01 (Crop and animal production, hunting and related services, 1.75%)
EU welfare increment	0.100 percentage points over Annex 2-A alone

Panel B. EU welfare sensitivity

EU input-price wedge	EU welfare	Increment over Annex 2-A
5%	0.053	0.049
10%	0.105	0.100
20%	0.216	0.211

Panel C. Illustrative 10 percent placement comparison

Region	EU input-price wedge inc.	ARG/BRA production-shock inc.
ARG	0.005	0.208
BRA	0.007	0.182
EU	0.100	0.001

Notes: Panel A reports an illustrative 10 percent EU access scenario. Panel B reports EU welfare changes in percent for 2022 under 5, 10, and 20 percent access-related input-cost changes. Panel C reports welfare increments over Annex 2-A alone. Welfare entries are percent deviations from the same-year benchmark CP equilibrium unless explicitly labeled as scenario increments. The REE channel is a maintained upstream-cost scenario using a rare-earth-related proxy bundle, not part of the Annex 2-A tariff estimand.

5.6 Aggregate Welfare and Synthesis

The aggregate welfare numbers confirm why the earlier sectoral and scenario results matter. Argentina has the largest endpoint gain in the earliest benchmark year, 0.031 percent in 2007, and falls to 0.009 percent in 2022. Brazil's gain is 0.024 percent in 2007 and 0.016 percent in 2022. The European Union remains around one-hundredth of a percent or less in every benchmark year and falls to 0.005 percent in 2022. These are percent changes, not decimal welfare changes: a value such as 0.05 means 0.05 percent, not 5 percent. The goods-tariff schedule therefore generates very small aggregate effects, including Argentina's 0.0009 percent loss in 2011.

The small magnitude reflects the narrower estimand used here relative to broader deep-agreement estimates such as [Timini and Viani \(2020\)](#). Their preferred structural-gravity exercise estimates the trade effect of agreements with

similar deep provisions and maps that effect into bilateral trade-cost reductions; the estimated shock implies roughly 60 percent higher North-to-South exports and 90 percent higher South-to-North exports before general-equilibrium feedback. This paper instead changes only the coded Annex 2-A goods-tariff schedule. Timini and Viani measure a full deep-agreement trade-cost shock, while the benchmark here measures the direct tariff component of the legal schedule.

The scale comparison makes clear that tariff removal alone is too small to reproduce deep-agreement effects. In 2022, the Annex 2-A benchmark gives welfare gains of 0.009 percent for Argentina, 0.016 percent for Brazil, and 0.005 percent for the European Union. A more aggressive internal-tariff benchmark, which caps EU–Argentina and EU–Brazil sector tariffs at the near-internal Argentina–Brazil margin, raises those values only to 0.022, 0.033, and 0.008 percent. Even the zero-tariff upper bound gives only 0.025, 0.037, and 0.009 percent. Table 9 reports the full comparison. The comparison indicates that reproducing deep-agreement magnitudes requires channels outside the coded goods-tariff schedule.

Table 9: Scale of Tariff-Only and Deep-Agreement Counterfactuals, 2022

Experiment	Object	ARG	BRA	EU
Annex 2-A benchmark	Legal goods-tariff schedule	0.009	0.016	0.005
Internal-tariff benchmark	Sector tariffs capped at the Argentina–Brazil internal maximum	0.022	0.033	0.008
Zero-tariff upper bound	Same bilateral goods tariffs set to zero	0.025	0.037	0.009
External deep-agreement estimate	Timini–Viani structural-gravity shock	0.300	0.400	0.070

Notes: The first three rows report this paper’s 2022 CP counterfactuals; for those rows, all entries are percent deviations from the same-year benchmark CP equilibrium unless explicitly labeled as scenario increments. The external deep-agreement estimate reports the Timini–Viani structural-gravity benchmark for Argentina, Brazil, and the average EU effect (Timini and Viani, 2020). It differs by shock, model, and sample and is included only as a scale comparison; it is not a CP tariff-only counterfactual.

Comparing Annex 2-A with other tariff-policy counterfactuals helps put the magnitudes in context. [Caliendo and Parro \(2015\)](#) study NAFTA tariff reductions and find much larger gains for Mexico, 1.31 percent, but small effects for the United States and Canada, 0.08 and -0.06 percent. [Ossa \(2014\)](#) studies global tariff bargaining and reports larger welfare effects because the counterfactual tariffs are much larger: trade-war tariffs average above 60 percent and the welfare loss from a breakdown of cooperation averages 2.9 percent. [Bagwell et al. \(2021\)](#) show that tariff-negotiation counterfactuals can also generate small aggregate magnitudes: their model predicts a 0.06 percent world real-income gain from Uruguay Round MFN tariff bargaining and a -0.02 percent world real-income effect under a no-MFN counterfactual. The current EU–Mercosur tariff component belongs to this tariff-only class, but it is especially small because the relevant applied tariff level and bilateral exposure are small after aggregation.

Table 10: Aggregate Welfare Effects of Annex 2-A

Year	Avg. effective cut	ARG	BRA	EU	CHN	USA	ROW
2007	32.6	0.031	0.024	0.007	0.000	0.000	0.000
2011	32.3	-0.001	0.027	0.009	0.000	0.000	0.000
2014	31.1	0.020	0.021	0.007	0.000	0.000	0.000
2017	30.7	0.019	0.021	0.007	0.000	0.000	0.000
2022	31.2	0.009	0.016	0.005	0.000	0.000	0.000

Notes: All welfare entries are percent deviations from the same-year benchmark CP equilibrium unless explicitly labeled as scenario increments. Avg. effective cut is the mean sector effective reduction, weighted by positive-value agreement-pair trade, including zero-tariff services. It is neither the mean legal cut nor the reduction in an average tariff in percentage points.

The results separate three objects with different empirical status: the legal tariff schedule, external-risk diagnostics, and maintained input-cost scenarios. First, applying the final signed legal schedule as a common counterfactual shock across benchmark-year economies produces small welfare changes, including Argentina’s small loss in 2011, concentrated Argentina–Brazil manufacturing losses, and a politically exposed EU agri-food margin. Second,

partner reallocation and U.S. tariff risk are diagnostics around that tariff schedule: the sourcing changes are CP counterfactual responses, and the U.S. tariff exercise is a uniform stress test rather than an observed policy path. Third, the rare-earth channel is a maintained input-cost scenario whose cross-country welfare pattern depends on whether the primitive is EU access or Mercosur upstream supply. The old sectoral resistance margins do not vanish, and the tariff schedule alone is not enough to explain larger claims about the agreement. What the results provide is narrower: they identify economic margins that make the agreement more defensible under the current environment, without estimating lobbying, ratification, or the probability of signature.

6 Conclusion

After two decades of delay, the EU–Mercosur tariff schedule is best read as a small direct tariff shock whose interpretation changes with the surrounding economic environment. The exercise supports a conditional answer. Annex 2-A is a small aggregate shock whose sectoral and reallocation effects vary when the signed schedule is introduced into different benchmark economies. The U.S. tariff-risk and rare-earth exercises then show how external tariff pressure and rare-earth input-cost exposure can change the interpretation of that tariff benchmark. The model does not identify the decisive bargaining event or estimate signature probabilities; it shows which economic margins make the agreement more defensible than a tariff-only reading would suggest.

The paper’s central result is that the tariff component cannot carry the renewed economic case by itself. Annex 2-A generates very small welfare changes across years, including a small loss for Argentina in 2011, and those changes do not scale mechanically with the average tariff cut because trade weights, partner composition, sector sizes, and input-output linkages change over time. Its main result is distributional: small aggregate gains coexist with concentrated manufacturing losses for Argentina and Brazil in metals, machinery, electrical equipment, chemicals, and transport-related sectors, while the EU agricultural margin appears only as a bloc-level agri-food diagnostic in this model.

The broader economic case appears only when the tariff schedule is read against changed sourcing, U.S. tariff risk, and rare-earth input-cost exposure. The zero-tariff full-liberalization and internal-tariff benchmarks show that more aggressive tariff-only shocks remain small, while larger effects in broader estimates should be interpreted as coming from channels outside the coded goods-tariff schedule. In the 2022 environment, China-plus-U.S. displacement is more relevant for Argentina and Brazil’s sourcing, U.S. tariff risk creates selective sectoral offsets, and rare-earth scenarios show how maintained input-cost changes can dominate the tariff component. The contribution is therefore a disciplined measurement exercise: the paper isolates the legal tariff schedule, then shows why external risk and rare-earth input-cost exposure can make the same agreement economically easier to defend in a different world.

References

- Al Jazeera and News Agencies (2019). Ireland, france set to block EU–Mercosur trade deal over amazon. News report. Published August 23, 2019. Available at <https://www.aljazeera.com/economy/2019/8/23/ireland-france-set-to-block-eu-mercotur-trade-deal-over-amazon>.
- Alfaro, L. and Chor, D. (2023). Global supply chains: The looming “great reallocation”. Working Paper 31661, National Bureau of Economic Research.
- Alfaro, L., Fadinger, H., Schymik, J. S., and Virananda, G. (2025). Trade and industrial policy in supply chains: Directed technological change in rare earths. Working Paper 33877, National Bureau of Economic Research. Revision date March 2026.
- Amiti, M., Redding, S. J., and Weinstein, D. E. (2020). Who’s paying for the us tariffs? a longer-term perspective. *AEA Papers and Proceedings*, 110:541–546.
- Associated Press (2024). French farmers escalate protests against an EU–Mercosur trade deal and fear unfair competition. News report. Published November 18, 2024. Available at <https://apnews.com/article/farmers-protest-france-mercotur-eu-trade-06174a5b46cb1b9ea69c5a6a4b153ac7>.
- Autor, D. H., Dorn, D., and Hanson, G. H. (2013). The china syndrome: Local labor market effects of import competition in the united states. *American Economic Review*, 103(6):2121–2168.
- Bagwell, K., Staiger, R. W., and Yurukoglu, A. (2021). Quantitative analysis of multiparty tariff negotiations. *Econometrica*, 89(4):1595–1631.
- Baqae, D. and Farhi, E. (2022). Networks, barriers, and trade. Working Paper 26108, National Bureau of Economic Research. Originally circulated July 2019; revised May 2022.
- Bosker, M. and Westbrock, B. (2024). The network origins of the gains from trade. *Journal of Economic Theory*, 216:105800.
- Caliendo, L. and Parro, F. (2010). Welfare gains from changing partners in a trade bloc: the case of mercosur. In *New Developments in Computable General Equilibrium Analysis for Trade Policy*. Emerald Group Publishing Limited.
- Caliendo, L. and Parro, F. (2015). Estimates of the trade and welfare effects of nafta. *The Review of Economic Studies*, 82(1):1–44.
- Campos, R. G. and Timini, J. (2022). Unequal trade, unequal gains: the heterogeneous impact of mercosur. *Applied Economics*, 54(49):5655–5669.
- Cavalcanti, T., Ogeda, P. M., and Ornelas, E. (2025). The us-china trade war creates jobs (elsewhere). Working paper, CESifo.
- Costa, F., Garred, J., and Pessoa, J. P. (2016). Winners and losers from a commodities-for-manufactures trade boom. *Journal of International Economics*, 102:50–69.
- Costinot, A. and Rodríguez-Clare, A. (2014). Trade theory with numbers: Quantifying the consequences of globalization. In *Handbook of international economics*, volume 4, pages 197–261. Elsevier.
- Council of the European Union (2026a). EU–Mercosur: Council greenlights safeguards for agricultural products. Press release. Published March 5, 2026. Available at <https://www.consilium.europa.eu/en/press/press-releases/2026/03/05/eu-mercotur-council-greenlights-safeguards-for-agricultural-products/>.
- Council of the European Union (2026b). EU–Mercosur: Council greenlights signature of the comprehensive partnership and trade agreement. Press release. Published January 9, 2026. Available at <https://www.consilium.europa.eu/en/press/press-releases/2026/01/09/eu-mercotur-council-greenlights-signature-of-the-comprehensive-partnership-and-trade-agreement/>.

- Dhingra, S., Huang, H., Ottaviano, G., Pessoa, J. P., Sampson, T., and Van Reenen, J. (2017). The costs and benefits of leaving the EU: Trade effects. *Economic Policy*, 32(92):651–705.
- Euronews (2026). Hundreds of tractors rumble through paris in protest over EU–Mercosur deal. News report. Published January 13, 2026. Available at <https://www.euronews.com/2026/01/13/hundreds-of-tractors-rumble-through-paris-in-protest-over-eu-mercoshur-deal>.
- European Commission (2026a). EU–Mercosur partnership agreement: Enhancing trade and investment in critical raw materials. Factsheet. Available at https://policy.trade.ec.europa.eu/eu-trade-relationships-country-and-region/countries-and-regions/mercoshur/eu-mercoshur-agreement/factsheet-eu-mercoshur-partnership-agreement-enhancing-trade-and-investment-critical-raw-materials_en.
- European Commission (2026b). EU–Mercosur: Text of the agreement. Official trade agreement page. The European Union and Mercosur signed the EMPA and iTA on January 17, 2026. Available at https://policy.trade.ec.europa.eu/eu-trade-relationships-country-and-region/countries-and-regions/mercoshur/eu-mercoshur-agreement/text-agreement_en.
- European Commission (2026c). The EU–Mercosur trade agreement. Official agreement explainer. Last updated March 23, 2026. Available at https://commission.europa.eu/topics/trade/eu-mercoshur-trade-agreement_en.
- European Commission (2026d). EU trade relations with Mercosur. Official trade relationship page. Available at https://policy.trade.ec.europa.eu/eu-trade-relationships-country-and-region/countries-and-regions/mercoshur_en.
- European Commission (2026e). European critical raw materials act. Policy page. Available at https://commission.europa.eu/topics/eu-competitiveness/green-deal-industrial-plan/europe-an-critical-raw-materials-act_en.
- European Commission, Directorate-General for Taxation and Customs Union (2026). Provisional application of the EU–Mercosur interim trade agreement begins 1 may 2026. Official news article. Published April 24, 2026. Available at https://taxation-customs.ec.europa.eu/news/provisional-application-eu-mercoshur-interim-trade-agreement-begins-1-may-2026-2026-04-24_en.
- European Union and Mercosur (2026a). EU–Mercosur interim trade agreement: Annex 2-a, tariff elimination schedules. Agreement annex. Official Annex 2-A tariff schedules accessed from the European Commission agreement-text page. Available at https://policy.trade.ec.europa.eu/eu-trade-relationships-country-and-region/countries-and-regions/mercoshur/eu-mercoshur-agreement/text-agreement_en.
- European Union and Mercosur (2026b). EU–Mercosur interim trade agreement: Annex 2-b, export duties. Agreement text. Available at <https://www.gov.br/siscomex/pt-br/acordos-comerciais/arquivos/2026/acordo-mercoshul-uniao-europeia/textos-ingles-acordo-ita/anexo-02-b-direitos-de-exportacao.pdf>.
- European Union and Mercosur (2026c). EU–Mercosur interim trade agreement: Chapter 2, trade in goods. Agreement text. Available at <https://www.gov.br/siscomex/pt-br/acordos-comerciais/arquivos/2026/acordo-mercoshul-uniao-europeia/textos-ingles-acordo-ita/02-comercio-de-bens.pdf>.
- Facchini, G., Silva, P., and Willmann, G. (2021). The political economy of preferential trade agreements: An empirical investigation. *The Economic Journal*, 131(640):3207–3240.
- Fajgelbaum, P. D., Goldberg, P. K., Kennedy, P. J., and Khandelwal, A. K. (2020). The return to protectionism. *Quarterly Journal of Economics*, 135(1):1–55.
- Federal Register (2025). Executive order 14257: Regulating imports with a reciprocal tariff to rectify trade

- practices that contribute to large and persistent annual united states goods trade deficits. Presidential document. Published April 7, 2025. Available at https://www.federalregister.gov/documents/full_text/html/2025/04/07/2025-06063.html.
- Gaulier, G. and Zignago, S. (2010). BACI: International trade database at the product-level. the 1994–2007 version. Working Paper 2010-23, CEPII.
- Government of Brazil (2024). Brazil’s federal government releases full text of partnership agreement between mercosur and the european union. Official government news release. Published December 10, 2024. Available at <https://www.gov.br/planalto/bd/en/latest-news/2024/12/brazil2019s-federal-government-releases-full-text-of-partnership-agreement-between-mercosur-and-the-european-union>.
- Grieger, G. (2020). Amazon deforestation and EU–Mercosur deal. European Parliamentary Research Service, At a Glance. Published October 29, 2020. Available at [https://www.europarl.europa.eu/thinktank/en/document/EPRS_ATA\(2020\)659311](https://www.europarl.europa.eu/thinktank/en/document/EPRS_ATA(2020)659311).
- Grossman, G. M. and Helpman, E. (1994). Protection for sale. *American Economic Review*, 84(4):833–850.
- Guimbard, H., Jean, S., Mimouni, M., and Pichot, X. (2012). Macmap-hs6 2007, an exhaustive and consistent measure of applied protection in 2007. *International Economics*, 130:99–121.
- International Energy Agency (2026). Rare earth elements. Report. Published April 8, 2026. Available at <https://www.iea.org/reports/rare-earth-elements>.
- Kemp, M. C. and Wan, H. Y. (1976). An elementary proposition concerning the formation of customs unions. *Journal of International Economics*, 6(1):95–97.
- Kharazi, A., Hopkinson, P., Zils, M., and Kobayashi, T. (2025). A dynamic production network approach to model resource productivity shocks from policy interventions in the uk. *International Economics and Economic Policy*, 22(63).
- Krishna, P. (1998). Regionalism and multilateralism: a political economy approach. *The quarterly journal of economics*, 113(1):227–251.
- Lipsey, R. G. (1957). The theory of customs unions: Trade diversion and welfare. *Economica*, 24(93):40–46.
- Liu, E. (2019). Industrial policies in production networks. *Quarterly Journal of Economics*, 134(4):1883–1948.
- Meade, J. E. (1955). The theory of customs unions. (*No Title*).
- MERCOSUR (2024). Joint statement by the mercosur signatory states and the european commission on the conclusion of the association agreement negotiations. Official joint statement. Published December 6, 2024. Available at <https://www.mercosur.int/en/declaracion-conjunta-mercosur-union-europea>.
- Mesquita Moreira, M., Rodríguez Chatruc, M., Lage de Sousa, F., and Merchán, F. (2020). Trade, productivity, innovation, and employment: Lessons from the impact of chinese competition on manufacturing in brazil. Technical report, Inter-American Development Bank.
- Organisation for Economic Co-operation and Development (2023). Oecd inter-country input-output tables. Database. Available at <https://www.oecd.org/sti/ind/inter-country-input-output-tables.htm>.
- Ossa, R. (2014). Trade wars and trade talks with data. *American Economic Review*, 104(12):4104–4146.
- Ossa, R. (2016). Quantitative models of commercial policy. In *Handbook of commercial policy*, volume 1, pages 207–259. Elsevier.
- Paz, L. S. (2018). The effect of import competition on brazil’s manufacturing labor market in the 2000s: Are imports from china different? *The International Trade Journal*, 32(1):76–99.
- Riezman, R. (1985). Customs unions and the core. *Journal of International Economics*, 19(3-4):355–365.
- Santacreu, A. M., Sposi, M., and Zhang, J. (2023). What determines state heterogeneity in response to us tariff changes? Working paper, Federal Reserve Bank of Chicago.

- Teti, F. (2024). Missing tariffs. Working Paper 11590, CESifo. Available at SSRN: <https://ssrn.com/abstract=5097020>.
- The White House (2025a). Fact sheet: President donald j. trump modifies the scope of reciprocal tariffs and establishes procedures for implementing trade deals. Fact sheet. Published September 5, 2025. Available at <https://www.whitehouse.gov/fact-sheets/2025/09/fact-sheet-president-donald-j-trump-modifies-the-scope-of-reciprocal-tariffs-and-establishes-procedures-for-implementing-trade-deals/>.
- The White House (2025b). Further modifying the reciprocal tariff rates. Executive order. Published July 31, 2025. Available at <https://www.whitehouse.gov/presidential-actions/2025/07/further-modifying-the-reciprocal-tariff-rates/>.
- Timini, J. and Viani, F. (2020). A highway across the atlantic? trade and welfare effects of the EU-Mercosur agreement. Technical Report 2023, Banco de Espana.
- U.S. Geological Survey (2025). Mineral commodity summaries 2025: Rare earths. Technical report, U.S. Geological Survey. Available at <https://pubs.usgs.gov/periodicals/mcs2025/mcs2025.pdf>.
- Viner, J. (1982). The economics of customs unions. In *International Economics Policies and their Theoretical Foundations*, pages 163–164. Elsevier.
- World Bank and UNCTAD (2026). World integrated trade solution: Unctad trains tariff data. Database. Available at <https://wits.worldbank.org/>.
- World Trade Organization (2014). China – measures related to the exportation of rare earths, tungsten and molybdenum. Dispute settlement summary, DS431/DS432/DS433. Available at https://www.wto.org/english/tratop_e/dispu_e/cases_e/ds432_e.htm.

Appendix

The appendix records the material needed to reproduce and interpret the quantitative exercises without repeating the main Results section. Appendix 7 documents the agreement documentation and tariff mapping. Appendix 8 documents the data, calibration, and exercise files. Appendix 9 keeps only secondary results needed to read the political and decomposition margins. Appendices 10, 11, and 11.1 report the U.S. tariff-risk, rare-earth input-cost, and zero-tariff full-liberalization robustness exercises behind the four measured margins.

7 Agreement Text, Tariff Extraction, and Model Mapping

This appendix records how the legal agreement text is converted into the tariff shock used in the quantitative exercise. The main distinction is between provisions that enter the model as tariff changes and provisions that matter institutionally but remain outside the counterfactual.

7.1 Source Recovery and Source Index

The agreement files used in the paper come from the European Commission agreement-text page and are stored in `sources/eu_mercosur_agreement/official_files/`. The recovered trade and legal bundle contains 31 nonzero PDFs. Annex 2-A, the tariff-elimination schedule, has 2320 pages and readable text according to `pdfinfo` and `pdftotext`.

The extraction workbook in `sources/eu_mercosur_agreement/` stores the source index, legal extraction, normalized HS6 schedule, sector mapping, sensitive products, random text audit, and validation checks.

7.2 Agreement Provisions Relevant to the Paper

Annex 2-A defines the goods-tariff shock that enters the quantitative model. The remaining provisions define implementation conditions, market-access rules, and political constraints that matter for interpretation but are not converted into model shocks.

Table 11: Agreement provisions used to define the paper’s counterfactual scope

Agreement part	What it does	Use in this paper
Annex 2-A	Sets product-level tariff elimination, staging, exclusions, tariff-rate quotas, and special treatments.	Main modeled shock.
Chapter 3	Defines when a shipment is originating and can claim the preference.	Relevant implementation condition, not modeled as a cost.
Chapters 5 and 6	Keep technical, food-safety, animal-health, and plant-health controls in force while adding co-operation and procedures.	Relevant for agricultural politics and market access, not modeled as a cost.
Chapter 9 and Annex 9-A	Allow temporary suspension of preferences under injury conditions; Annex 9-A adds vehicle-specific treatment.	Flags why some lines are not simple permanent tariff cuts.
Chapter 12	Opens covered public tenders subject to listed entities, thresholds, and exclusions.	Institutional gain, not in the tariff counterfactual.
Chapter 18 and Annex 18-A	Adds commitments on labour, climate, biodiversity, forests, and deforestation.	Central to ratification politics, not in the tariff counterfactual.
Chapters 21 and 22	Define consultation, dispute settlement, and committees that supervise implementation.	Governance condition for effectiveness, not a model shock.

7.3 Legal Tariff Schedule

The goods schedule is Annex 2-A. It is expressed at the legal tariff-line level: CN 2013 for EU imports from Mercosur and NCM 2012 for Mercosur imports from the European Union. The extraction script reads Annex 2-A with `pdftotext -tsv`, uses page coordinates to recover table columns, and stores one row per legal tariff line. The raw extraction contains 19,406 legal rows: 9,376 in the EU schedule and 10,030 in the Mercosur schedule.

Each extracted row keeps the legal code, HS6 prefix, product description, importing side, exporting side, legal base-rate fields, staging category, source PDF page, legal page, and flags for quotas, safeguards, exclusions, and special treatment when these can be inferred from the schedule. A deterministic 25-row audit compares sampled legal codes and staging categories against the extracted PDF page text; all sampled rows pass this text-based audit.

Table 12: Agreement Extraction Validation

Check	Value	Status	Interpretation
Recovered nonzero agreement PDFs	31	OK	Selected trade and legal bundle.
Tariff-schedule PDF pages	2,320	OK	Readable by <code>pdftotext</code> and <code>pdftotext</code> .
Raw legal tariff rows	19,406	OK	Rows parsed from the EU and Mercosur tariff tables.
Normalized schedule rows	49,496	OK	Mercosur legal rows expanded by importing member where base rates differ.
Invalid HS6 codes	0	OK	HS6 prefix recovered from legal CN/NCM codes.
Duplicate importer-exporter-HS6-stage rows	0	OK	No duplicate normalized legal-stage cells.
Rows in deterministic text audit	25	OK	Sampled legal codes and staging categories checked against page text.

This table audits extraction from the agreement documentation only. The separate model-coverage diagnostic below explains which valid HS6 goods codes fall outside the current BACI-based model object.

The extraction is complete at the HS6 level, but the model uses only HS6 products that also appear in the project's BACI-based sector concordance and tariff-weight object. Table 13 separates these two steps. Products outside the current model object are still Annex 2-A merchandise lines, not non-tradable sectors and not goods omitted from the agreement. They are retained in the legal audit trail, but they receive zero model weight when they do not appear in the relevant BACI tariff-weight object. Their model treatment follows data-support constraints rather than legal exclusion from Annex 2-A.

Table 13: Legal HS6 Coverage and Use in the Model Object

Diagnostic	Count	Interpretation
Unique legal HS6 goods products recovered from the agreement documentation	5,206	The legal extraction has complete HS6 product identifiers: there are no missing or invalid HS6 codes in the parsed tariff schedule.
Unique legal HS6 goods products already in the project model concordance	4,090	These goods products are directly linked to the sector concordance used by the 49-sector model.
Unique Annex 2-A HS6 goods products without current BACI model support	1,116	These are covered legal goods lines, not non-tradables and not goods outside the agreement. They lack model support because the project concordance is built from the BACI HS92 product universe used for tariff weights.
Legal schedule rows attached to those HS6 products	11,837	The row count is larger than the product count because CN/NCM legal lines and Mercosur member rows map to the same HS6 product.
Rows for modeled importers: EU, Argentina, and Brazil	7,005	These legal rows would affect the modeled EU–Argentina/Brazil goods shock only if they had sector and trade-weight support.
Rows for Paraguay and Uruguay	4,832	These rows remain in the legal audit trail but are outside the current six-region counterfactual shock.
Outside-object HS6 products appearing in relevant BACI tariff weights	0	None of the missing HS6 products is present in the BACI HS6 weight file used to aggregate the modeled bilateral goods shock.
Positive-trade BACI rows with missing HS6 products	0	The missing-concordance products therefore receive zero weight in the current model rather than being assigned to an arbitrary sector.

The largest missing chapters by unique HS6 count are Fish HS03 (136); Organic chemicals HS29 (100); Electrical machinery HS85 (68); Iron and steel HS72 (53); Machinery HS84 (45); Miscellaneous chemical products HS38 (40); Wood HS44 (36); Knitted fabrics HS60 (36). These codes are legal Annex 2-A goods products, not services, non-tradables, or goods omitted from the agreement. Their sector tariff effect is zero in the current counterfactual because they have no relevant BACI tariff-weight support. No downloaded HS-revision concordance is applied here; the model concordance is generated locally from the BACI HS6 products used in the project and the hand-coded HS2-to-ICIO map in `scripts/sector_concordance_utils.R`.

Table 14 classifies the Annex 2-A products without current BACI model support by the same HS2-to-ICIO logic used elsewhere in the project. The table is diagnostic only: it describes what the products are economically, but it does not force them into the counterfactual shock.

Table 14: Goods Sectors of Annex 2-A HS6 Products Without Current Model Support

Inferred goods sector	Unique HS6	Legal rows	Modeled-importer rows	Largest HS chapters
Heavy Manufacturing	290	3,052	1,796	Electrical machinery HS85 (68); Iron and steel HS72 (53); Machinery HS84 (45)
Agriculture	273	2,025	1,257	Fish HS03 (136); Fruit and nuts HS08 (27); Live animals HS01 (26)
Light Manufacturing	223	2,254	1,334	Wood HS44 (36); Knitted fabrics HS60 (36); Hides and leather HS41 (32)
Chemicals and Plastics	200	3,354	1,824	Organic chemicals HS29 (100); Miscellaneous chemical products HS38 (40); Rubber HS40 (29)
Food and Beverages	101	840	596	Meat HS02 (26); Prepared meat and fish HS16 (19); Prepared vegetables/fruit HS20 (18)
Mining and Energy	20	239	153	Mineral products HS25 (8); Ores HS26 (7); Mineral fuels HS27 (5)
Transport Equipment	9	73	45	Aircraft and spacecraft HS88 (4); Vehicles HS87 (3); Ships and boats HS89 (2)

Sectors are inferred only for diagnosis by applying the paper’s HS2-to-ICIO mapping to the Annex 2-A HS6 codes without current BACI model support. The inference shows these are tradable merchandise categories concentrated in heavy manufacturing, agriculture, light manufacturing, chemicals, and food. They are not used in the tariff shock because they have no support in the relevant BACI tariff-weight object, not because they are absent from the agreement.

7.4 From Legal Lines to Model Sectors

Let k denote an HS6 product and let ℓ denote legal 8-digit CN or NCM lines inside that HS6 product. The extraction first assigns a simple final legal cut share c_ℓ to each legal line:

$$c_\ell = \begin{cases} 1, & \text{if the line has a standard final duty-free staging category,} \\ 0, & \text{if the line is excluded or has no simple ad valorem final cut,} \\ p, & \text{if the text gives a simple partial preference } p \in (0, 1). \end{cases}$$

This rule is conservative. Tariff-rate quotas, specific-duty formulas, entry-price treatments, and other non-simple legal treatments are flagged in the workbook and treated as zero simple ad valorem cuts unless a simple final cut share is explicitly recoverable.

When several legal lines map to the same importer–exporter–HS6 cell, the model input uses the average legal cut share across those legal lines:

$$c_{ni,k} = \frac{1}{L_{ni,k}} \sum_{\ell \in \mathcal{L}_{ni,k}} c_\ell.$$

This equal weighting across legal tariff lines is a coding assumption. Trade weighting enters in the next step. The mean legal cut uses BACI weights:

$$\bar{c}_{ni,t}^j = \sum_{k \in \mathcal{H}(j)} w_{ni,k,t} c_{ni,k}.$$

The counterfactual tariff applies each product's cut before aggregation:

$$\tau_{ni,t}^{j,A} = \sum_{k \in \mathcal{H}(j)} w_{ni,k,t} \tau_{ni,k,t}^B (1 - c_{ni,k}),$$

where $\tau_{ni,k,t}^B$ is the year-specific HS6 tariff from the MACMAP-based benchmark. The effective sector cut is $C_{ni,t}^j = 1 - \tau_{ni,t}^{j,A} / \tau_{ni,t}^{j,B}$ for a positive baseline tariff and zero otherwise; it need not equal $\bar{c}_{ni,t}^j$.

7.5 Scope of the Counterfactual

The counterfactual includes bilateral goods tariff reductions between the European Union and Argentina and between the European Union and Brazil. It does not model services, procurement, rules-of-origin compliance, SPS or TBT compliance costs, quota rents, within-quota and out-of-quota allocation, safeguards as state-contingent policy, sustainable-development enforcement, or dispute settlement. Those provisions are part of the institutional framework, but they are not converted into shocks in the current quantitative exercise.

7.6 Agreement Content, Affected Sectors, and Exceptions

The legal schedule is broad, but non-simple treatments are concentrated in food, agriculture, transport equipment, and selected manufacturing groups. This explains why the coded agreement schedule produces smaller effects than the separate zero-tariff full-liberalization upper bound and why politically sensitive margins remain even when aggregate gains are positive.

Table 15: Agreement treatment by model sector group

Direction	Sector group	Rows	Full elim. (%)	Quota rows	Excluded rows
Argentina/Brazil imports from EU	Agriculture	748	97.1	2	20
Argentina/Brazil imports from EU	Chemicals and Plastics	5,244	94.5	0	288
Argentina/Brazil imports from EU	Food and Beverages	744	83.3	56	66
Argentina/Brazil imports from EU	Heavy Manufacturing	5,212	90.7	0	486
Argentina/Brazil imports from EU	Light Manufacturing	2,552	84.5	0	396
Argentina/Brazil imports from EU	Mining and Energy	326	100.0	0	0
Argentina/Brazil imports from EU	Transport Equipment	402	74.1	20	82
EU imports from Mercosur	Agriculture	573	91.8	37	8
EU imports from Mercosur	Chemicals and Plastics	1,158	99.3	8	0
EU imports from Mercosur	Food and Beverages	1,026	72.6	163	102
EU imports from Mercosur	Heavy Manufacturing	2,394	100.0	0	0
EU imports from Mercosur	Light Manufacturing	1,637	100.0	0	0
EU imports from Mercosur	Mining and Energy	168	100.0	0	0
EU imports from Mercosur	Transport Equipment	247	100.0	0	0

Rows are legal schedule rows used by the model-side extraction. Full elimination refers to the final legal cut share coded as one. Quota and exclusion rows are kept as legal restrictions and are not forced into ad valorem equivalents.

Table 16 records the product groups that matter most for interpreting the agreement in the paper. On the EU side, the main managed concessions are in agricultural and food products, especially beef, poultry, sugar, ethanol, rice, honey, and related products. On the Mercosur side, the main sensitive margins include dairy, garlic, automobiles, and selected manufacturing lines.

Table 16: Protected, excluded, and specially treated product groups

Side	Products	Legal treatment	Relevance for the paper
EU imports from Mercosur	Beef and poultry	Tariff-rate quotas and product-specific treatment; out-of-quota trade keeps the base duty.	Keeps the most visible EU agricultural opposition inside the institutional section but outside a simple tariff-cut interpretation.
EU imports from Mercosur	Ethanol, sugar, rice, honey	Tariff-rate quotas or partial preferences; ethanol includes a chemical-use reservation.	Shows why the legal schedule protects sensitive agricultural and energy-related products.
EU imports from Mercosur	Sheep/goat meat, selected dairy/sugar lines	Excluded rows or rows with no simple final ad valorem cut in the extracted schedule.	Explains why EU-side liberalization is not full free trade in sensitive food products.
EU imports from Mercosur	Dairy, eggs, garlic, starches, biodiesel	Tariff-rate quotas and staged preferences.	Shows that some EU concessions are managed quantities rather than pure tariff cuts.
Mercosur imports from EU	Milk powder, cheese, infant formula, garlic	Tariff-rate quotas with staged in-quota preferences.	Identifies Mercosur-side limits on EU agri-food market access.
Mercosur imports from EU	Vehicles and new-technology vehicles	Long staging and vehicle-specific safeguard or special treatment.	Connects the agreement to Mercosur manufacturing and auto-policy concerns.
Mercosur imports from EU	Selected fish, dairy, chemicals, machinery, metals, and light manufactures	Excluded rows concentrated in manufacturing and some food lines.	Explains why the Annex 2-A scenario is below the zero-tariff robustness case.

7.7 Rare-Earth-Related Products

Rare-earth products enter through the ordinary goods-tariff schedule when the relevant HS6 products are matched to Annex 2-A. The rare-earth exercise treats those products separately as input-cost exposure while keeping the legal coding within the ordinary tariff schedule.

Table 17: Rare-earth-related HS6 products in Annex 2-A

Direction	HS6	Product coverage in the legal schedule	Legal base rate	Treatment in the tariff schedule
Argentina/Brazil imports from EU	253090	Mineral substances n.e.s.; HS6 includes rare-earth ores and concentrates	4.0%	Standard tariff elimination
Argentina/Brazil imports from EU	280530	Rare-earth metals, scandium and yttrium	2.0%	Standard tariff elimination
Argentina/Brazil imports from EU	284610	Cerium compounds	2.0%	Standard tariff elimination
Argentina/Brazil imports from EU	284690	Other rare-earth compounds	2.0–10.0%	Standard tariff elimination
EU imports from Mercosur	253090	Mineral substances n.e.s.; HS6 includes rare-earth ores and concentrates	Free	Standard tariff elimination
EU imports from Mercosur	280530	Rare-earth metals, scandium and yttrium	2.7–5.5%	Standard tariff elimination
EU imports from Mercosur	284610	Cerium compounds	3.2%	Standard tariff elimination
EU imports from Mercosur	284690	Other rare-earth compounds	3.2%	Standard tariff elimination

The rare-earth bundle is not a special chapter or carve-out in the agreement. These HS6 lines enter, when matched, through the ordinary goods schedule. HS6 253090 is broader than rare-earth ores; the project treats it as a proxy because the rare-earth legal detail sits below HS6.

8 Data Construction, Sector Aggregation, and Calibration Details

This appendix records the data choices behind Section 4. The main text keeps only the inputs needed to understand the counterfactual. The tables below document the sector aggregation, tariff completion rule, elasticity calibration, and benchmark objects used in the exercises.

Table 18: Appendix Reproducibility Map

Object	Main script or source	Key output	Appendix use
Agreement extraction	Source-recovery and tariff-extraction scripts	HS6 schedule; extraction workbook	Legal schedule, validation, and model mapping in A.1
Agreement appendix assets	Agreement-appendix script	Agreement appendix asset folder	Agreement provisions, treatment counts, sensitive products, rare-earth legal lines
Baseline model exercise	Calibrated same-agreement script	Calibrated exercise output folder	Welfare, sector effects, and decomposition checks
Results assets	Results asset script	Results-section asset folder	Compact main-text figures and appendix cross-checks
U.S. tariff exercise	U.S. tariff counterfactual script	U.S. tariff output and asset folders	External-policy robustness in A.4
Rare-earth extension	Rare-earth full-exercise script	Rare-earth output and asset folders	Strategic-input access and China-risk diagnostics in A.5
Zero-tariff upper bound	Three-scenario appendix script	Three-scenario output and asset folders	Legal schedule versus full liberalization in A.6

Exact paths are stored in the corresponding `scripts/`, `data_clean/`, and `reports/` folders; the table reports the reproducibility chain rather than every file path.

Table 19: Scenario Parameters Used in Appendix Exercises

Scenario	Shock	Applied to	Role in the paper
Coded agreement schedule	Sector tariff becomes $\sum_{k \in \mathcal{H}(j)} w_{ni,k,t} \tau_{ni,k,t}^B (1 - c_{ni,k})$ for matched EU–Argentina and EU–Brazil goods flows	All benchmark years	Main counterfactual; holds the legal schedule fixed while the benchmark economy changes
Zero-tariff upper bound	Same bilateral goods pairs set to zero tariffs	All benchmark years	Measures the distance between the coded legal schedule and unrestricted bilateral goods liberalization
U.S. tariff +10pp	Uniform 10 percentage-point increase in U.S. goods-import tariffs against non-U.S. origins	2022 benchmark	Main external-policy environment exercise
U.S. tariff +25pp	Uniform 25 percentage-point increase in U.S. goods-import tariffs against non-U.S. origins	2022 benchmark	High-intensity robustness case
EU rare-earth access	10 percent reduction in the EU rare-earth input price index; Argentina and Brazil keep base-line access	2022 benchmark	Strategic-input access extension
China-risk case	China-risk rare-earth trade-cost change with Mercosur access maintained	2022 benchmark	Diagnostic for whether agreement access offsets strategic-input risk
All-importer China exposure	China-risk cost change mechanically applied through each importer’s China-origin rare-earth shares	2022 benchmark	Diagnostic only; not the preferred mechanism

The agreement and zero-tariff scenarios use the calibrated five-year benchmark objects. The U.S. tariff and rare-earth extensions use the same 2022 benchmark and add only the scenario-specific channels described in A.4 and A.5.

8.1 Sector Aggregation

The model is solved in the 49-sector ICIO system. The 8-sector grouping is used only for exposition, broad heatmaps, and robustness summaries. HS6 products are first mapped to the 49-sector system and then to the 8-sector reporting groups.

Table 20: Sector Aggregation Used for Reporting

Broad sector	49-sector count	HS6 count	49-sector codes
Agriculture	3	388	A01, A02, A03
Food and Beverages	1	309	C10T12
Mining and Energy	8	150	B05, B06, B07, B08, B09, C19, D, E
Chemicals and Plastics	3	946	C20, C21, C22
Light Manufacturing	4	1326	C13T15, C16, C17_18, C31T33
Heavy Manufacturing	7	1752	C23, C24A, C24B, C25, C26, C27, C28
Transport Equipment	3	132	C29, C301, C302T309
Other	20	0	F, G, H49, H50, H51, H52, H53, I, J58T60, J61, J62_63, K, L, M, N, O, P, Q, R, S

Notes: The 49-sector codes are the sectors used in the model. HS6 counts refer to goods products mapped to each reporting group. The services and non-goods activities collected in ‘Other’ do not receive HS6 tariff lines directly.

8.2 Tariff Completion and Aggregation

MACMAP tariffs are completed at HS6 before they are aggregated to model sectors. If a tariff is missing at the importer–exporter–HS6–year level, the build searches in this order: the same importer–exporter–year within HS4; the same importer–year–HS6 across exporters; the same importer–year–HS4 across exporters; the same importer–year–sector; and finally zero if no tariff information is available in those cells. This hierarchy keeps product and partner information where it exists and avoids treating every missing raw tariff as free trade. Across direction–year cells, legal schedule coverage ranges from 63.52 to 86.81 percent of retained HS6 trade value, while imputed tariffs cover 8.85 to 84.03 percent. These are coverage limits on the retained product–weight object, not claims of complete legal or tariff support.

After completion, HS6 tariffs are aggregated with BACI trade weights:

$$\tau_{ni,t}^{j,B} = \sum_{k \in \mathcal{H}(j)} w_{ni,k,t} \tau_{ni,k,t}^B.$$

The same HS6-to-sector map is used to aggregate post-agreement tariffs after applying the legal Annex 2-A cut to each HS6 baseline tariff. Legal lines that cannot be converted into a simple final ad valorem cut are flagged in the agreement workbook and treated conservatively in the benchmark tariff shock, as described in Appendix 7.

8.3 Elasticities

The reported benchmark uses the promoted five-year WITS hierarchy. Accepted WITS estimates enter without shrinkage, and unsupported sector-years remain at the mapped Caliendo–Parro value. The low count of strict direct sectors reflects the acceptance rule rather than a sector drop: the 49-sector model contains service and other non-HS6 sectors, and among goods sectors the narrow sector-year estimate is used only when the country-level CP regression is positive, precise, and supported by enough products and country triples. The promoted WITS files are:

```
data_clean/calibration/theta_wits_country_cp_benchmark_final.csv
data_clean/calibration/theta_cpstyle_yearly_final.csv
```

The five-year WITS panel makes it possible to use broader estimating samples without returning to the earlier penalty method. The rule follows the treatment of weak or unavailable elasticities in the quantitative trade literature. [Caliendo and Parro \(2015\)](#) estimate sectoral elasticities with the CP triple-ratio equation but do not mechanically accept unstable estimates; when signs are not robust across samples, they use a manufacturing mean. [Ossa \(2016\)](#) emphasizes that tariff-elasticity estimation is often fragile and that CP-style estimates can fail when discriminatory tariff variation is limited, which supports pooling and sensitivity analysis rather than accepting weak cells. [Costinot and Rodríguez-Clare \(2014\)](#) use CP sectoral elasticities for goods sectors while assigning services an external aggregate value, and [Bagwell et al. \(2021\)](#) distinguish traded from nontraded sectors when using WITS/TRAINS tariff data in a quantitative tariff model. [Dhingra et al. \(2017\)](#) likewise combine HS tariff aggregation with external service elasticities and report robustness to alternative elasticity values.

The hierarchy is therefore a no-penalty sequence. Let $\mathcal{T} = \{2007, 2011, 2014, 2017, 2022\}$ and let $b(j)$ denote the broad goods group containing ICIO sector j . For each sector-year, the promoted value is

$$\theta_{j,t}^{hier} = \begin{cases} \hat{\theta}_{j,t}^{direct}, & \text{if the strict sector-year WITS estimate passes;} \\ \hat{\theta}_{j,\mathcal{T}}^{sector}, & \text{if the same sector pooled across all benchmark years passes;} \\ \hat{\theta}_{b(j),t}^{group}, & \text{if the broad goods group in the same year passes;} \\ \hat{\theta}_{b(j),\mathcal{T}}^{group}, & \text{if the broad goods group pooled across years passes;} \\ \theta_j^{static}, & \text{otherwise.} \end{cases}$$

The pooled regressions still use HS6-level WITS/TRAINS observations and the same CP triple-ratio equation. Pooling changes the estimating sample; it does not shrink a noisy WITS estimate toward the CP value. A value is accepted only when the corresponding sample passes the same sign, support, and precision checks used by the strict direct estimator. Services, non-HS sectors, and goods sectors that still fail after pooling keep the mapped Caliendo–Parro value.

Table 21 summarizes the resulting five-year hierarchy. The hierarchy raises WITS-supported sector-year cells from 34 strict direct cells to 75 WITS-based cells before the CP fallback, and the current result tables are rebuilt under this vector. Table 22 summarizes the promoted 2022 distribution by broad reporting group. Table 23 reports selected 49-sector elasticities that are central for the paper’s mechanisms: agriculture and food, the mining and chemicals sectors used in the rare-earth proxy, and the manufacturing sectors used for the political-margin discussion.

Table 21: Promoted Five-Year WITS Hierarchy

Year	Sectors	Direct	Same-sector pooled	Group-year	Group pooled	CP fallback
2007	49	1	9	3	2	34
2011	49	1	10	0	1	37
2014	49	9	2	1	1	36
2017	49	9	3	2	1	34
2022	49	14	1	5	0	29
All cells	245	34	25	11	5	170

Notes: Direct counts strict sector-year WITS estimates. Same-sector pooled counts cells filled by the same ICIO sector estimated across all five benchmark years. Group-year and group pooled use the broad goods group in the same year or across all years. CP fallback keeps the mapped Caliendo–Parro value. The table describes the promoted hierarchy used in the reported results.

Table 22: Elasticity Distribution by Broad Sector in 2022

Broad sector	Sectors	WITS based	θ_{min}	θ_{med}	θ_{max}
Agriculture	3	2	1.12	1.15	9.11
Food and Beverages	1	1	1.13	1.13	1.13
Mining and Energy	8	0	8.22	13.53	64.85
Chemicals and Plastics	3	3	0.41	1.65	2.87
Light Manufacturing	4	4	1.00	2.31	4.31
Heavy Manufacturing	7	7	0.91	2.19	22.12
Transport Equipment	3	3	1.41	1.82	2.22
Other	20	0	8.22	8.22	8.22

Notes: ‘WITS based’ counts sectors whose 2022 elasticity is supported by WITS at one of the hierarchy levels rather than kept at the mapped Caliendo–Parro value.

Table 23: Selected Sectoral Elasticities Used in the Main Mechanisms

Sector	Broad sector	2007	2011	2014	2017	2022	2022 source
A01	Agriculture	0.66	0.66	2.15	0.66	1.12	wits_direct_observed_trim99
C10T12	Food and Beverages	0.88	0.88	1.17	1.49	1.13	wits_direct_observed_trim99
B08	Mining and Energy	17.23	13.53	13.53	13.53	13.53	cp_static_fallback
C20	Chemicals and Plastics	3.13	3.13	3.13	3.13	0.41	wits_direct_trim99
C24A	Heavy Manufacturing	23.72	23.33	12.20	37.88	22.12	wits_direct_observed_trim99
C24B	Heavy Manufacturing	2.66	2.66	1.85	2.50	5.12	wits_direct_observed_trim99
C25	Heavy Manufacturing	6.99	6.99	6.99	1.76	2.19	wits_direct_observed_trim99
C27	Heavy Manufacturing	12.91	12.91	12.91	12.91	1.76	wits_sector8_year_observed_trim99
C28	Heavy Manufacturing	1.45	1.45	1.45	1.45	0.91	wits_direct_observed_trim99
C29	Transport Equipment	0.99	0.99	0.95	1.98	1.41	wits_direct_observed_trim99
C301	Transport Equipment	1.22	1.22	1.22	2.10	1.82	wits_sector8_year_observed_trim99
C302T309	Transport Equipment	2.22	2.22	2.22	2.22	2.22	wits_sector_pooled_trim99

Notes: The selected sectors cover the main model margins discussed in the paper. Source labels indicate whether the final value is the mapped static value or a supported WITS hierarchy estimate.

8.4 Benchmark Objects

The final benchmark objects combine the ICIO production network, completed and aggregated tariffs, and the promoted WITS no-penalty hierarchy elasticity vector. Table 24 reports the object dimensions and the solver check used before running the counterfactual exercises.

Table 24: Calibrated Benchmark Objects

Year	Regions	Sectors	Trade rows	θ rows	Solver check
2007	6	49	1764	49	pass
2011	6	49	1764	49	pass
2014	6	49	1764	49	pass
2017	6	49	1764	49	pass
2022	6	49	1764	49	pass

Notes: Trade rows equal importer–exporter–sector cells in the six-region, 49-sector system. The solver check records the no-shock smoke test for each calibrated benchmark year.

9 Additional Main-Exercise Results

This appendix keeps only the secondary results behind Section 5: descriptive partner-share changes, the aggregate welfare decomposition, and the institutional record behind the EU agricultural sensitivity discussed in the main text.

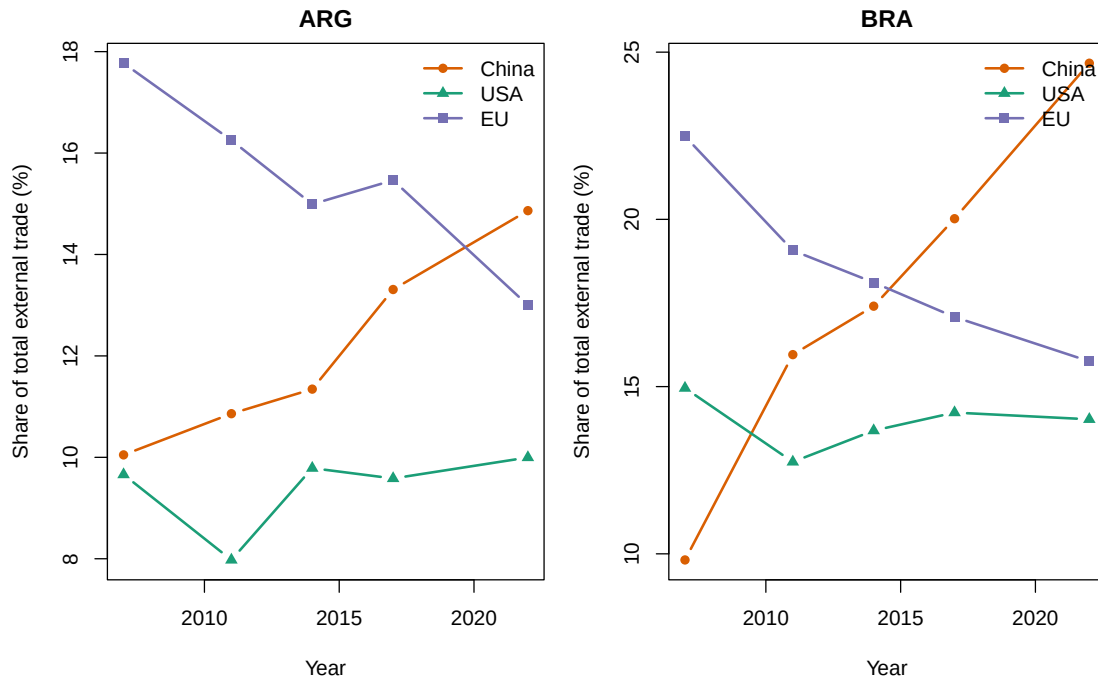


Figure 3: Descriptive Partner Shares in Mercosur External Trade

Notes: The figure reports baseline partner shares in total external trade for Argentina and Brazil. These are descriptive shares from the benchmark data, not counterfactual outcomes.

Table 25: Welfare Decomposition for Argentina, Brazil, and the European Union

Year	Region	Welfare (%)	Terms of trade	Volume of trade
2007	ARG	0.031	-0.073	0.104
2007	BRA	0.024	-0.038	0.061
2007	EU	0.007	0.005	0.002
2011	ARG	-0.001	-0.018	0.017
2011	BRA	0.027	-0.023	0.050
2011	EU	0.009	0.005	0.003
2014	ARG	0.020	-0.030	0.050
2014	BRA	0.021	-0.021	0.042
2014	EU	0.007	0.005	0.002
2017	ARG	0.019	-0.022	0.041
2017	BRA	0.021	-0.024	0.045
2017	EU	0.007	0.005	0.002
2022	ARG	0.009	-0.019	0.028
2022	BRA	0.016	-0.024	0.041
2022	EU	0.005	0.004	0.001

Table 26: EU Agricultural Sensitivity in the Institutional Record

Country or bloc	Exposed products	Evidence used
France	Beef and poultry; dairy often mentioned as a possible offset	French opposition is centered on agriculture; sensitive quotas named as beef and poultry
Ireland	Sensitive agricultural products, in particular beef	Irish government statement flags concern about sensitive agricultural products, in particular beef
Poland	Poultry first, then beef, pork, sugar, ethanol, cereals	European Parliament question on Polish agriculture names poultry as the main concern
Romania	Poultry, sugar, ethanol, beef	European Parliament 2019 question cites Romanian concern and lists poultry, sugar, ethanol and beef
EU safeguard list	Beef, poultry, dairy, sugar, ethanol	Consilium safeguard regulation names the sensitive products monitored by the EU

10 U.S. Tariff-Risk Scenario

The main Results section uses one core U.S. tariff-risk scenario: a uniform 10 percentage-point increase in U.S. goods-import tariffs against all non-U.S. origins. This appendix reports the high-intensity 25 percentage-point robustness case. The shock is a six-region approximation motivated by U.S. reciprocal-tariff policy; it does not reproduce the full HTS schedule, product exceptions, retaliation, or bilateral negotiations ([Federal Register, 2025](#); [The White House, 2025b,a](#)).

$$\tau_{USA,i,2022}^{j,U} = \tau_{USA,i,2022}^{j,B} + \Delta\tau^{US}, \quad i \neq USA, \quad j \in \mathcal{G},$$

where $\mathcal{G}$ is the set of goods sectors. The main text uses $\Delta\tau^{US} = 0.10$; the robustness case uses $\Delta\tau^{US} = 0.25$. In the 2022 scope check, the shock applies to 27 goods sectors and adds the same number of tariff points to the baseline U.S. goods-import tariff against every non-U.S. origin in the six-region system.

Table 27: Incremental Annex 2-A Effect Under Alternative U.S. Tariff Environments

Region	Annex 2-A	Annex 2-A under U.S. +10pp	Shift	Annex 2-A under U.S. +25pp	Shift
ARG	0.0094	0.0090	-0.0003	0.0088	-0.0005
BRA	0.0161	0.0135	-0.0025	0.0125	-0.0035
CHN	0.0003	0.0004	0.0000	0.0004	0.0000
EU	0.0048	0.0048	0.0000	0.0048	0.0000
USA	-0.0003	-0.0001	0.0001	0.0000	0.0003
ROW	-0.0001	-0.0001	0.0000	-0.0001	0.0000

11 Rare-Earth Access, Production, and Risk Extension

This appendix records the rare-earth extension used at the end of the results section. The extension keeps the main Annex 2-A tariff shock fixed and adds one maintained upstream-input channel to the same Caliendo–Parro solver in two placements: a direct EU input-price change and an Argentina–Brazil upstream production-cost change. The preferred interpretation of the EU input-price change is export-risk insurance. The goods chapter and Annex 2-B discipline export duties, export licensing, export monopolies, quantitative restrictions, and export-price requirements, so a lower EU input-price change is interpreted as reduced exposure to export-tax or export-restriction price premia ([European Union and Mercosur, 2026c,b](#); [European Commission, 2026a](#)). The motivation is narrow. The WTO rare-earth dispute documents the trade-policy concern around export restrictions ([World Trade Organization, 2014](#)). [Baqae and Farhi \(2022\)](#) provides the production-network logic for propagation,

while Alfaro et al. (2025) links rare-earth restrictions to input-output channels. I use Liu (2019) as related evidence on industrial-network propagation. I use Bosker and Westbrock (2024) and Kharazi et al. (2025) only to place the exercise near the broader literature on network structure and supply-chain risk; this appendix treats the mechanism as a partial counterfactual input-cost change.

The rare-earth bundle uses HS6 products 253090, 280530, 284610, and 284690. In the 49-sector model, HS6 253090 maps into mining sector B08, while the metal and compound lines map into chemicals sector C20. For importer n , source i , and year t , the source share in the rare-earth bundle is

$$\omega_{ni,t}^{REE} = \frac{M_{ni,t}^{REE}}{\sum_h M_{nh,t}^{REE}}.$$

Given a scenario-specific rare-earth delivered input-price change $\hat{p}_{ni,t}^{REE}$, the rare-earth input price index is

$$\hat{P}_{n,t}^{REE} = \left[\sum_i \omega_{ni,t}^{REE} \left(\hat{p}_{ni,t}^{REE} \right)^{-\theta_{REE}} \right]^{-1/\theta_{REE}}.$$

Here θ_{REE} is a source-substitution elasticity inside the rare-earth price index, not a production elasticity between rare earths and other inputs. The code sets it equal to the calibrated C20 trade elasticity from the same CP object, because the narrow rare-earth metal and compound lines map to C20. The broader B08/C20 weighted value is saved as a diagnostic, but it is not the central parameter because HS6 253090 is a broad ore proxy. The preferred direct EU access case sets $\hat{P}_{EU,t}^{REE} = 0.90$ directly, with sensitivity cases at 0.95 and 0.80, so the primitive access-insurance price change does not depend on θ_{REE} . The 10 percent number is a maintained avoided price premium, not an estimate of the agreement's legal effect on export duties.

Sector exposure comes from ICIO input-cost shares. Let $s_{nj,t}^{B08}$ and $s_{nj,t}^{C20}$ denote the shares of mining and chemicals inputs in sector j in region n , and let ω_t^{B08} and ω_t^{C20} denote the year-specific shares of those upstream sectors in the rare-earth bundle. The exposure term is

$$\rho_{n,j,t}^{REE} = s_{nj,t}^{B08} \omega_t^{B08} + s_{nj,t}^{C20} \omega_t^{C20}.$$

The rare-earth channel changes the unit-cost term by $(\hat{P}_{n,t}^{REE})^{\rho_{n,j,t}^{REE}}$. In the code, this is implemented as an origin-sector delivered-cost change, multiplying every sale of origin n and sector j by $(\hat{P}_{n,t}^{REE})^{\rho_{n,j,t}^{REE}}$, including domestic sales.

The alternative production-shock placement moves the primitive from EU input prices to Argentina and Brazil's upstream productive capacity in the same rare-earth-mapped sectors. For source country o , importer n , sector j , and year t , the delivered-cost multiplier is

$$\hat{d}_{ni,t}^{j,RAB} = (1 + g)^{-\omega_{j,t}} \quad \text{if } i \in \{ARG, BRA\} \text{ and } j \in \{B08, C20\},$$

and $\hat{d}_{ni,t}^{j,RAB} = 1$ otherwise. The preferred case sets $g = 0.10$, with sensitivity cases $g \in \{0.05, 0.20\}$. The weights $\omega_{j,t}$ are the year-specific B08 and C20 shares in the rare-earth bundle, so the shock is larger in B08 when rare-earth ores dominate the bundle and larger in C20 when metals and compounds dominate. Output is not imposed mechanically; the production-cost change affects Argentina and Brazil's delivered costs for domestic and foreign sales, and the CP equilibrium determines output, trade shares, prices, and welfare.

Table 28: Rare-Earth Production-Shock Placement in 2022

Panel A. Primitive-to-outcome map

Primitive/object	Production-shock case
REE bundle definition	HS6 253090, 280530, 284610, and 284690
Sector mapping	HS6 253090 → B08 (Other mining and quarrying); HS6 280530, 284610, and 284690 → C20 (Chemicals and chemical products)
Shocked origins	Argentina and Brazil
Preferred primitive shock	+10% effective REE productive capacity, with 5% and 20% sensitivity cases
Model implementation	Exporter-sector productivity wedge for Argentina/Brazil B08 and C20, implemented as an origin-sector delivered-cost reduction for domestic and foreign sales
2022 bundle weights	B08: 70.14%; C20: 29.86%
Implied broad-sector cost changes	B08: -6.466%; C20: -2.806%
Top affected sectors	BRA B06 (2.815 pp); ARG B06 (1.901 pp); ARG B09 (1.393 pp); BRA B08 (1.320 pp); BRA B09 (1.190 pp)
EU welfare increment	0.001 percentage points over Annex 2-A alone

Panel B. Sensitivity to the production shock

Shock	ARG welfare	ARG inc.	BRA welfare	BRA inc.	EU welfare	EU inc.
5%	0.117	0.107	0.110	0.094	0.005	0.000
10%	0.218	0.208	0.198	0.182	0.006	0.001
20%	0.402	0.392	0.362	0.346	0.006	0.002

Notes: Panel A reports the preferred 10 percent production-shock placement. Panel B reports welfare changes and increments over Annex 2-A alone for 5, 10, and 20 percent production shocks. The production shock uses the same HS6 bundle and B08/C20 mapping as the EU input-price change.

Following [Baqae and Farhi \(2022\)](#), the diagnostic also computes the direct-plus-indirect input-output exposure implied by the ICIO cost-share matrix. Let $\Gamma_{n,t}(k, j)$ be the cost share of input sector k in using sector j . The network diagnostic is

$$\rho_{n,t}^{net} = (I - \Gamma'_{n,t})^{-1} \rho_{n,t}.$$

This object is reported only to interpret the footprint of the primitive shock. It is not used in the counterfactual input-cost change because the CP solver already propagates $(\hat{P}_{n,t}^{REE})^{\rho_{n,t}}$ through downstream prices, quantities, and welfare. Replacing $\rho_{nj,t}$ with $\rho_{nj,t}^{net}$ inside the input-cost change would double count the network channel.

Table 29: Direct and Network Rare-Earth Exposure in the EU, 2022

Code	Sector	Direct (%)	Network (%)	Mult.
C20	Chemicals and chemical products	7.60	10.98	1.45
C22	Rubber and plastic products	5.97	9.66	1.62
C23	Other non-metallic mineral products	4.60	6.57	1.43
B08	Other mining and quarrying	3.50	4.50	1.29
C17_18	Paper products and printing	1.56	3.53	2.26
C13T15	Textiles, wearing apparel, leather and related products	1.61	3.48	2.16
A01	Crop and animal production, hunting and related services	1.75	3.29	1.88
C21	Pharmaceuticals, medicinal chemical and botanical products	1.57	2.88	1.83

Notes: Direct exposure is $\rho_{nj,t}$, the cost share of B08/C20 rare-earth-linked inputs. Network exposure is $(I - \Gamma'_{n,t})^{-1} \rho_{n,t}$ using the EU 2022 ICIO input-cost matrix. The multiplier is network exposure divided by direct exposure. The table is a diagnostic only; the model input-cost change uses direct exposure.

The main text focuses on EU rare-earth access, with 5, 10, and 20 percent sensitivity around the preferred 10 percent avoided price premium, and adds a compact comparison with the Argentina–Brazil production-shock placement. The remaining cases below are robustness exercises: China risk with maintained Mercosur access, the combined access-plus-risk scenario, and an all-importer China exposure diagnostic.

Table 30: Rare-Earth Extension: Welfare Effects in 2022

Scenario	Argentina	Brazil	European Union
Observed baseline	0.000	0.000	0.000
EU REE access only, 5 pct	0.002	0.003	0.049
EU REE access only, 10 pct	0.005	0.007	0.100
EU REE access only, 20 pct	0.011	0.015	0.211
China risk with Mercosur access only	-0.011	-0.017	-0.040
China import-exposure diagnostic only	-0.049	-0.113	-0.040
Annex 2-A only	0.009	0.016	0.005
Annex 2-A + EU REE access, 5 pct	0.012	0.020	0.053
Annex 2-A + EU REE access, 10 pct	0.014	0.023	0.105
Annex 2-A + EU REE access, 20 pct	0.020	0.031	0.216
Annex 2-A + China risk with Mercosur access	-0.002	-0.001	-0.035
Annex 2-A + China import-exposure diagnostic	-0.039	-0.096	-0.035
Annex 2-A + EU REE access + China risk	0.004	0.007	0.065

Table 31: Rare-Earth Extension: China-Risk Mechanisms in 2022

Region	Main risk REE price (%)	Import-exposure price (%)	Main risk welfare	Import-exposure welfare
ARG	0.000	1.715	-0.002	-0.039
BRA	0.000	4.407	-0.001	-0.096
EU	3.350	3.350	-0.035	-0.035

Table 31 separates the preferred China-risk case from the diagnostic. In the preferred case, the rare-earth price index rises for the European Union but not directly for Argentina or Brazil. In the all-importer exposure diagnostic, Argentina and Brazil are mechanically hit through their China-origin rare-earth import shares.

11.1 Agreement Schedule and Zero-Tariff Upper Bound

This appendix reports three versions of the tariff experiment. The first scenario is the benchmark-year baseline: observed trade and tariffs in each year, with no counterfactual tariff change. The second scenario is the coded agreement schedule used in the main exercise. It applies each extracted HS6 legal cut to its year-specific MACMAP tariff before aggregation, so the counterfactual sector tariff is $\tau_{ni,t,j}^A = \sum_{k \in \mathcal{H}(j)} w_{ni,k,t} \tau_{ni,k,t}^B (1 - c_{ni,k})$. The third scenario is a zero-tariff robustness check. It uses the same bilateral country pairs and model sectors, but sets the EU–Argentina, EU–Brazil, Argentina–EU, and Brazil–EU goods tariffs to zero.

The comparison therefore separates three objects: the baseline level of protection in each benchmark year, the legal schedule implied by the agreement text, and a full-liberalization upper-bound. The coded agreement and zero-tariff experiments are both applied to the same benchmark-year economies. They are not simulations of the phase-in path after ratification.

Table 32: Coded Agreement Schedule and Zero-Tariff Upper Bound

Year	Cut EU (%)	Cut MS (%)	ARG welfare		BRA welfare		EU welfare	
			Agreement	Zero	Agreement	Zero	Agreement	Zero
2007	22.6	42.8	0.031	0.101	0.024	0.063	0.007	0.016
2011	20.9	40.5	-0.001	0.025	0.027	0.064	0.009	0.018
2014	18.3	38.3	0.020	0.054	0.021	0.044	0.007	0.014
2017	21.3	36.1	0.019	0.044	0.021	0.042	0.007	0.013
2022	16.8	40.7	0.009	0.025	0.016	0.037	0.005	0.009

Cut EU is the trade-weighted mean sector effective cut on EU imports from Argentina and Brazil; Cut MS is the trade-weighted mean sector effective cut on Argentina and Brazil imports from the EU. The cut averages include positive-value agreement-pair cells, including zero-tariff services; they are not tariff-point reductions. Welfare entries are percent changes relative to each benchmark-year baseline. The zero-tariff columns set the same bilateral goods tariffs to zero and are an upper-bound comparison, not a phase-in scenario.

The zero-tariff case is useful because it shows the distance between the legally coded agreement and an unrestricted liberalization exercise. Across years, the Annex 2-A welfare changes are smaller than the zero-tariff gains; Argentina has a small loss under Annex 2-A in 2011. In the calibrated exercise, the Annex 2-A welfare changes average 0.28 of the zero-tariff gains for Argentina, 0.44 for Brazil, and 0.51 for the European Union.